Charles University

Faculty of Social Sciences
Institute of Economic Studies

MASTER'S THESIS

Reinforcement learning in Agent-based macroeconomic model

Author: **Bc. Bořivoj Vlk**
Supervisor: **PhDr. Ing. Jiří Skuhrovec, PhD.**
Academic Year: **2017/2018**

Declaration of Authorship

The author hereby declares that he compiled this thesis independently, using only the listed resources and literature, and the thesis has not been used to obtain a different or the same degree.

The author grants to Charles University permission to reproduce and to distribute copies of this thesis document in whole or in part.

Prague, July 31, 2018

Signature

Acknowledgments

The author is grateful to PhDr. Ing. Jiří Skuhrovec PhD. for his comments and remarks on both author's bachelor's and this thesis.

Abstract

Utilizing game theory, learning automata and reinforcement learning concepts, thesis presents a computational model (simulation) based on general equilibrium theory and classical monetary model.

Model is based on interacting Constructively Rational agents. Constructive Rationality has been introduced in current literature as machine learning based concept that allows relaxing assumptions on modeled economic agents information and expectations.

Model experiences periodical endogenous crises (Fall in both production and consumption accompanied with rise in unemployment rate). Crises are caused by firms and households adopting to a change in price and wage levels. Price and wage level adjustments are necessary for the goods and labor market to clear in the presence of technological growth.

Finally, model has good theoretical background and large potential for further development. Also, general properties of games of learning entities are examined, with special focus on sudden changes (shocks) in the game and behavior of game's players, during recovery from which rigidities can emerge.

JEL Classification	D80, D83, C63, E32, C73,
Keywords	Learning, Information and Knowledge, Agent-based, Reinforcement learning, Business cycle, Stochastic and Dynamic Games, Simulation, Modeling

Author's e-mail	borivoj.vlk@gmail.com
Supervisor's e-mail	jskuhrovec@gmail.com

Abstrakt

Za využití teoretický konceptů z teorie her, učících se automatonů a zpětnovazebního učení, práce prezentuje počítačový model (simulaci) založený na všeobecné rovnováze a učebnicovém dynamickém makroekonomickém modelu.

Model je založen na interagujících Konstruktivně Racionální agentech. Konstruktivní Racionalita je pojmem soudobé literatury, popisujícím rozhodování ekonomických agentů založeném na strojovém učení, který umožňuje uvolnit předpoklady a požadavky modelu na informace, kterými agenti disponují a správnost jejich očekávání.

V modelu se periodicky opakují období krizí, charakterizovných jako strmý propad úrovně produkce doprovázený stejně strmým nárůstem nezaměstnanosti. Příčinou krizí je změna v úrovně cen a mezd, na jejihž novou hladinu se firmy a domácnosti v modelu musí adaptovat. Změna úrovně cen a mezd je nezbytná pro zachování rovnováhy na trhu zboží pokud dochází k technologickému pokroku (růstu).

Model má dobré teoretické základy a velký potenciál pro další vývoj. Práce se rovněž zabývá obecnými vlastnostmi her vícero interagujících a učících se entit. Zvláštní důraz je kladen na náhlé změny ve hře a chování hráčů při těchto změnách, během kterých se objevují endogenní rigidity.

Klasifikace JEL D80, D83, C63, E32, C73,

Klíčová slova Učení, informace, Multiagentní, Zpětnovazební učení, Hospodářeský cyklus, Stochastické a dynamické hry, Simulace, Modelování

E-mail autora `borivoj.vlk@gmail.com`

E-mail vedoucího práce `jskuhrovec@gmail.com`

List of Figures

List of Tables

Contents

1 Introduction

Recent financial crisis impacted not only the real economy, but as it was after the Great depression, economic theory as well. One of the main issue that modern central bankers and macro economist had to tackle were financial frictions. New Keynesian DSGE models, in their comprehensiveness, were often found to lack many of the properties of the real economy and therefore failed to explain recent events. Many argued, that the next step DSGE should take is to introduce financial frictions into these models, which should enable them to explain (and hopefully predict) future crises, which will emerge from financial sector. (Maršál 2012) DSGE models that include frictions are often based on framework by Gilchrist 1999 or Moore 1997 and were often extended after recent financial crisis.

This thesis follows development of usage of reinforcement learning algorithms in economics like Marimon, McGrattan, and Sargent 1990 and Sinitskaya 2015. It aims to model frictions and rigidities endogenously, as a result of agent's decision making mechanism.

Thesis develops a macroeconomic simulation based on textbook, so called *classical monetary model* as presented in Galí 2008. Even designing such model, presets challenges when economic agents decision making mechanism is reinforcement learning algorithm. Thesis shall examine whether it is even possible. What assumptions can be dropped compared to original model and on the other hand, what new assumptions need to be introduced. After the model will be specified a computer simulation will be coded according to that specification. Results of the simulation will be examined in order to answer whether any endogenous cycles or rigidities emerged in the model.

Thesis has following structure:

- Section 2 introduces the reader into the concept of reinforcement learning
- Section 3 presets basic textbook economic model on which will later chapters build
- Section 4 shows connection of reinforcement learning to conventional optimization methods used in textbook economic models
- Section 5 is a summary on game theory concepts and its relationship to the model which thesis aims to develop
- Section 6 presets an example of multi-agent game with reinforcement learning and examines it for rigidities
- Sections 7 and 8 present the definition of the simulation and its results

1.1 Literature

This thesis combines several fields, core literature utilized in this thesis is reviewed in this section.

1.1.1 Reinforcement learning

Core piece on reinforcement learning is most recent edition of Richard Bellman's Dynamic programming. (Bellman 2010). Dynamic programming was developed as a mathematical concept. Later reinforcement learning algorithm are based on Bellman's optimality principle. Work on particular reinforcement learning algorithm, called *Q-learning*, by Watkins 1989 and more importantly convergence proof provided in Dayan 1992 are used extensively in the technical specification of the model this thesis develops. A variant of Q-learning particularly useful in games of multiple learning players, called Nash Q-learning, is presented in (Hu and Wellman 2003).

1.1.2 Agent-based approach and games

Sinitskaya 2015 tested range of computer learning algorithms on macroeconomic decision problems and inspired the author to pursue the goal of applying such algorithm. Schwartz 2014 combines reinforcement learning and game theory and is a textbook which summarizes how learning agents interact in games. Judd 2006 provides wide range of topics on agent-based modeling.

1.1.3 Behavioral topics

Considerable amount of work on behavioral biases and judgment in economics was carried out by Kahneman 1974 and Tversky 1979. Their work, along with later Kahneman 2011 helps to establish a reasoning behind using some decision making method other than analytical dynamic optimization with complete knowledge of the environment. Namely a concept of heuristic rules and simplifications employed when making decision in complex environment or with incomplete information. Niv 2009 even provides more specific links between human and animal decision making on biological level and reinforcement learning. It is worth mentioning, that there are similarities between computer algorithm and biological processes in human and animal brains, although any conclusions based on this would be highly speculative. This was also addressed by Fellous 2010 and Ljungberg 1993 provides some relevant laboratory experiments. Collaborative work of Montague 1997 is a fusion of theories of adaptive optimization control and behavioral laboratory experiments.

1.1.4 Business cycle

Main source on macroeconomic modeling utilized by this thesis is Galí 2008. This textbook provides a classical monetary model of which an agent based variant with reinforcement learning is developed. Wide range of post-crisis literature on financial crisis was surveyed by Maršál 2012. Moore 1997 modeled amplification and persistence of shocks to technology.

2 Short guide to *Reinforcement learning*

Computer science develops different forms of "artificial intelligence" algorithms. Common goal of such algorithms is usually the ability of the computer system to artificially adopt and to solve certain problems in different environments without the need for a computer programmer to modify the algorithm. Such algorithms are commonly referred to as *Machine learning*. Class of machine learning algorithms, that this thesis utilize is called *Reinforcement learning*.

In this section, I present an example which should help the reader understand basic principles behind *reinforcement learning*. I will also provide some basic concepts, that try to find a relationship between reinforcement learning and actual human decision making.
Consider two players in a game of tennis. Each of them has a very simple goal - to win the game. During the course of the game, they have very limited amount of time to make their decisions and are forced to adapt to the other player's strategy. This forbids them to make any detailed analysis of their opponents behavior and they are forced to make simple and quick decision - mostly based on heuristic rules. Those rules may form different strategies of play against their opponent strategy and players learn them during training and throughout their career. But they are always forced to choose the best response strategy quickly and without much planning analysis. But which strategy should they as a response to their opponents strategy choose?
Let's say, that as the game develops, one of the players suddenly decides to change her/his strategy - starts to playing chopped balls. The other one has to react a and has to react quickly. His decision making can be described by several concepts, one of which was described by Kahneman 2011. Decision making of any individual may consist of two so called "systems". First of which is quick and more heuristics based. The other requires more time, but is able to draw conclusions, analyze facts and predict outcomes of actions without actually executing those actions. Reinforcement Learning in computer science tries to mimic the first system. A player may expect, that each of the strategies that she/he may respond to "chopped balls" has some effectivity. The effectivity of playing strategy can be in this case easily measured -

by the amount of points this strategy may bring for the rest of the game. The player has certain beliefs about the expected number of points she/he may win if she/he executes executes it and will therefore choose the strategy with highest amount of expected points. Let's call a function, that assigns expected number of points to each available strategy a *Value function*.

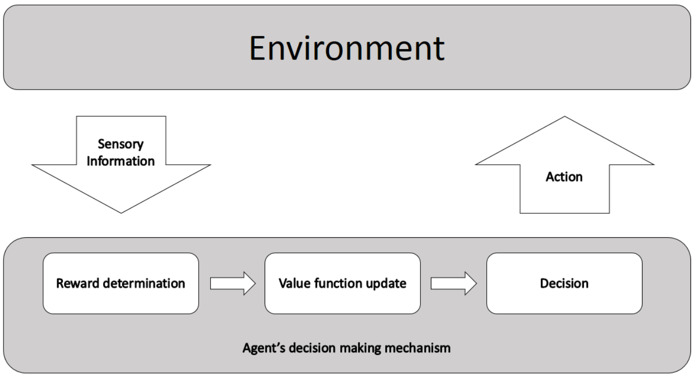

Figure 1: Reinforcement Learning diagram

Of course, our player may be wrong in her/his estimate of point particular strategy may bring her/him. If the actual number of points won is lower/larger than the expected amount, it is reasonable for the player to change/update the estimate and maybe choose different strategy next time. This procedure is summarized on figure 1. If the tennis game, including the opponent, forms the environment to which player reacts and reward would be measured in points obtained.
Successfulness of different tennis strategies may be measured in points, but one can find many cases, where the reward is unmeasurable. There is a wide range of literature, that documents, that similar procedure to the one depicted on the figure 1 indeed takes place inside animal and probably also human brain. Fellous 2010 Surprisingly enough, the activity of Dopaminergic neurons (dopamine-producing nerve cells) of measured subjects in laboratory experiments, so to say "mimics" the updating procedure of the *value function* estimate (Montague 1997). This was one of the reasons, why reinforcement learning was so popular during 90's. (Niv 2009) To better understand this relation, consider one of the probably most well known series of laboratory experiments on this topic carried out by Ljungberg 1993.

A monkey has been seated in front of three bulbs. Two of those are similar in shape and are referred to as indicator lights. Directly under each of those is a lever. The monkey can reach those levers from it's seating position. The third bulb, so called

trigger light, indicates when the monkey is supposed to press the lever assigned to each of the two indicator lights. Also, at the start of this experiment, monkey has to keep it's hand on the resting pad and cannot touch any of the two levers.
The monkey is trained to perform the following task: One of the indicator light comes on. Then the monkey has to wait with it's hand on the resting pad until trigger light comes on. Then, it has to reach and press the lever, that is under indicator light that came on previously.
If the monkey performs this task correctly, it will receive reward in form of apple juice. Also, there is a slight delay between the monkey pressing the correct lever and juice delivery.
During this experiment, activity of Dopaminergic(DA) neurons was monitored and the experiment was repeated several times. During the first trials of the experiment, when the monkey was supposed to press the lever, activity of DA neurons did not deviate from normal levels. But it increased dramatically, once the apple juice was delivered. Interestingly enough, as the experiment was repeated, activity of DA neurons after juice delivery declined to normal levels, but also increased above the normal level at the time, when the monkey was supposed to press the lever.
The experiment was carried out with several different setups and finally the researchers concluded, the the DA neurons activity level mirrors monkeys expectations about future rewards of its actions. (Montague 1997)

3 RBC model as *Markovian domain*

Agent-based model this thesis develops is based on "textbook" dynamic RBC model. Following section presents optimization problems faced by households and firms in Real Business Cycle (RBC) model, similar to the one described in chapter 2 of Gali(2008). Goal of this part is to state those optimization problems in form of Bellman's equation and define decision making mechanism of agents. In this section, "agent" refers to representative firm and/or household. Bellman's equation is condition of optimality in Dynamic programming on which reinforcement learning algorithm this thesis uses is based. Therefore, before we can define the model as stated in the introduction to this thesis, baseline RBC model needs to be presented and examined from Dynamic programming perspective.

RBC models was selected because it lacks any rigidities and non-neutrality of money and thesis shows, how those properties can arise in such a model based solely on agent's decision making mechanism.

3.1 Classical RBC model

Classical dynamic Real Business Cycle (RBC) framework is a model of economy, that assumes representative household and representative firm solving dynamic (time-dependent) optimization problem. Generally, model assumes discrete time. Both households and firms are price/wage takers. Model also assumes that Efficient Market Hypothesis (EMH) holds and crucially - Perfect Foresight Strong-Form Rational Expectations.
Strong-Form Rational expectations are a necessity for deriving model solution analytically - like presented in Galí 2008. Strong-Form Rational Expectations state, that all agents make optimal use of any information they have to form their expectations as accurately as possible. On top of that, they have all the information there is. Namely (following Tesfatsion 2016):

- Equations, variable classification, and admissibility conditions for the model, including the actual decision rules used by any other agent (private or public) appearing in the model to generate their actions and/or expectations.
- The true values for all deterministic exogenous variables for the model.
- All properties of the probability distributions governing stochastic exogenous variables that are known by the modeler at the beginning of period t.
- Realized values for all endogenous variables and stochastic exogenous variables as observed by the modeler through the beginning of period t.

Perfect Foresight Strong-Form rational expectations assumed in chapter 2 of Galí 2008 are an extension of Strong-Form Rational Expectations and are defined as follows (Tesfatsion 2016):

- Agent has Strong-Form Rational Expectations
- Agent's expectations in each period t are correct without error.

No one in the real world, not even economists in the central banks, has a set of information that would form at least Strong-Form Rational Expectations.
Strong-Form Rational Expectations are a necessary assumption for RBC and New-Keynesian DSGE models. Performance of models with such assumptions has been found insufficient during recent financial crisis. To solve this issue, other assumption - mainly financial frictions - have been introduced. This thesis takes another approach. It alters classical dynamic RBC model by using different decision making mechanism of agents.

3.2 Household's optimization problem

In "textbook" dynamic RBC mode, agents, namely households and firms, solve dynamic optimization problem. This optimization problem is hereby stated and later it is shown, how it can be formulated as *Markov decision process*.

Households want to maximize their expected lifetime utility. Assuming that households live infinitely, and this is a reasonable assumption(multiple generation and will of parents to provide heritage to their children), representative household solves following dynamic optimization problem:

$$\max_{\{C_t,N_t\}_{t=1}^{\infty}} E\sum_{t=0}^{\infty}\beta^t U(C_t,N_t)$$

Where C_t is quantity consumed of single good and N_t denoted hours of work or employment. The utility function $U(C_t,N_t)$ has the following properties in consumption domain:

$$\frac{\partial U(C_t,N_t)}{\partial C_t} > 0 \quad \text{and} \quad \frac{\partial^2 U(C_t,N_t)}{\partial C_t^2} \leq 0$$

For the hours-worked, following applies:

$$\frac{\partial U(C_t,N_t)}{\partial N_t} \leq 0 \quad \text{and} \quad \frac{\partial^2 U(C_t,N_t)}{\partial N_t^2} \leq 0$$

In each period, household faces budget constraints. Therefore, the optimization problem needs to be solved subject to:

$$P_tC_t + Q_tB_t \leq B_{t-1} + W_tN_t$$

Where P_t is price of consumption good. W_t denotes nominal wage and B_t represents the quantity of one-period , nominally riskless discount bonds purchased in period t and maturing in period $t+1$. Each bond pays one unit of money at maturity and it's price is Q_t. Household's present value of assets has to be non-negative and it's decision making is bound by "No Ponzi-game" restriction:

$$\lim_{T\to\infty} E_t\{B_T\} \geq 0$$

3.3 Firm's optimization problem

Firm seeks to maximize its profits each period t by producing goods Y_t, hiring labor N_t for cost of wage W_t and selling goods at price P_t. Firm's production function is given by:

$$Y_t = A_tN_t^{1-\alpha}$$

Where N_t denotes firms labor demand and A_t is some stochastic technology process that forms Markov chain.
As firm maximizes its profits in each period t it solves following optimization problem:

$$\max_{\{N_t\}_{t=1}^{\infty}} E \sum_{t=1}^{\infty} \beta^t (P_t Y_t - W_t N_t)$$

Were price P_t and wage W_t are given and therefore are a state variables. The amount of labor hired N_t is a control variable.

3.4 State and control in RBC model

Assuming that both firms and households are price and wage takers, variables P_t and W_t are state variables. From both firm's and household's perspective, they can follow some stochastic process. Amount of labor supplied/demanded by particular representative household/firm, denoted N_t is a household's/firm's control variable.
Let's assume, that there is a finite number of possible values of N_t. This assumption can be justified by the fact, that in real economy, there is a finite number of households and each of them can supply finite amount of labor and is willing to work only for some exact number of hours, therefore $N_t < \infty$ and $N_t \in \mathbb{N}_0$. Given the similar reasoning, the set of all possible values of B_t is assumed to be finite and $B_t \in \mathbb{N}_0$. For household, a combination of values of N_t and B_t will be for the rest of this thesis referred to as "action" and will be denoted a. A set of all possible a forms a set of actions, denoted $\mathcal{A}$. Also a set of all possible values of P_t, W_t and Q_t will be assumed to be finite. This can be justified by the fact, that wage and price is usually paid in some currency, which has a minimal nominal value. A combination of values of P_t, W_t and Q_t will be for the rest of this thesis referred to as "state" and will be denoted s. A set of all possible s is also finite and will be denoted $\mathcal{S}$.
For the firm, similar reasoning applies.
Both representative household and representative firm can, based on state and their action, with probability $P_a(s, s')$ receive some reward, which will be denoted $R_a(s, s')$. Meaning, reward received, given state s, action a in time t and resulting state s' in time $t+1$. Also For household, reward is utility and is specified by households utility function. For firm, reward is profit.
Above definitions and reasoning allow us to specify the model as a *stochastic Markov Decision process*.

3.4.1 stochastic Markov Decision Process

Throughout this thesis, *stochastic Markov Decision process*. is defined a tuple (S, A, P, R, β) where

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is finite set of actions
- $P_a(s, s\prime)$ is the probability that action a in the state s at time t will lead to state $s\prime$ in time $t+1$
- $R_a(s, s\prime)$ is the immediate utility (or reward) received after transition from state s to state $s\prime$
- $\beta \in [0, 1])$ is the discount factor, which represents the difference in importance between future rewards and preset rewards.

Based on the above definition and addition assumptions made in preceding subsection, we can state, that state of the economy forms a *Markov Chain* and both households and firms face a *stochastic Markov Decision process*.

3.5 Solving dynamic optimization problems with Dynamic programming

Real world households and firms usually don't solve the whole problem analytical in advance. Instead, they make their decision consequently in each period, while taking into account the future prospects of their decisions today. Reason behind this is, that often lack the ability to properly judge complicated risky alternatives against each other. (Tversky 1979)
Of course, households and firms (especially) plan their decision in advance, but they also face uncertainty and limited information about the future development of prices and wages. Since those variables are not defined in advance, usual analytical solution would be not well defined. Mathematically, process of finding the solution to dynamic optimization problem that is revoked (possibly revisited) in each period, can be captured using Bellman's *Dynamic programming* (Bellman 2010). Given usual assumption summarized for example in Galí 2008.Optimal values of control variable, i.e. solution of the optimization problem, for both households and firms, can be found using *Dynamic programing*. In the next section, optimization problem faced by households and firms will be stated as *Bellman's equation* which stands at the core of this method.
Let π be a set of **ordered** decisions a, $\pi = \{a_{s_0}, a_{s_1}, a_{s_2}, \ldots\}$. In this case, ordered means, that particular action is executed as the system/model evolves and is assigned to it's current state. Such set π will be for the rest of the thesis referred to as *policy*. This definition can be utilized in reformulating household's/firm's ultimate goal - household/firm seeks profit maximizing *policy*. Meaning, that household/firm

seeks such set of decisions, that will maximize its expected reward(utility/profit) over time,

$$\max_{\pi \in \mathcal{M}} \sum_{t=t_0}^{\infty} \sum_{s' \in \mathcal{S}} \beta^{t-t_0} R_{a_t}(s, s') * P_{a_t}(s, s')$$

Where $\mathcal{M}$ denotes a set of all possible *policies*. Solution of this problem will be denoted π^* and referred to as *optimal policy*.

3.5.1 Value function of *Dynamic programming*

The largest possible expected reward - i.e. the the expected reward, given that firm/household uses *optimal policy*, written as a function of the state, is called the *Value function*:

$$V(s) = \max_{\pi \in \mathcal{M}} \sum_{t=t_0}^{\infty} \sum_{s' \in \mathcal{S}} \beta^{t-t_0} R_{a_t}(s, s') * P_{a_t}(s, s')$$

One of the core ideas behind *Dynamic programming* is, that finding the *Value function* is equivalent to finding the optimal policy - solution to the optimization problem. The value function can be found by *Bellman's Optimality Principle*

An optimal policy has the property, that whatever the initial state and initial decisions are, the remaining decision must constitute an optimal policy with regard to the state resulting from the first decision.

This principle can be stated mathematically, as a recursive definition of *Value function*.

$$V(s) = \max_{a} \{R_a(s, s') + \beta E[V(s')]\}$$

This formal definition of *Bellman's Optimality Principle* is in the literature referred to as ***Bellman's equation*** and will form theoretical backbone of Reinforcement learning numerical solution method of dynamic macroeconomic models this thesis uses.

4 Solving *Bellman's equation* using *Reinforcement learning*

There are several way how to obtain a solution of *Bellman's equation*. One way to do it, is to use *Reinforcement learning* computer algorithm. Such algorithm, called

Q-Learning will be presented in this section. This algorithm has been chosen for its convergent property under standard assumptions of modern dynamic macroeconomic models. Under those assumptions, the algorithm is guaranteed to solve the model - i.e. finds *optimal policy*. On top of that, some of those assumptions - mainly strong-Form rational expectations can be relaxed. Behavior of the model under those weaker assumption can be then observed. This section follows closely Mitchell 1997

4.1 Q-Learning

Agent, which in this can can be either household or firm has a set of action $\mathcal{A}$ and faces a set of possible states of the world $\mathcal{S}$. Since time is discrete, at each step t agent senses the state of the world $s \in \mathcal{S}$ and chooses from action $a \in \mathcal{A}$. Agent's task is to find optimal policy, i.e. policy that maximizes expected reward. Formally, agent's task is to find π^*, such that:

$$\pi^* = \arg\max_{\pi \in \mathcal{M}} \sum_{t=t_0}^{\infty} \sum_{s' \in \mathcal{S}} \beta^{t-t_0} R_{a_t}(s, s') * P_{a_t}(s, s')$$

Let's assume, that the agent does not have the complete knowledge of probabilities $P_{a_t}(s, s')$, nor does he have complete knowledge of $R_{a_t}(s, s')$. Therefore, he can not learn the *optimal policy* directly - using for example some analytical method. The lack of knowledge about probabilities $P_a(s, s')$ and rewards $R_a(s, s')$ prevents agent from any attempt to evaluating rewards from possible actions in current state and therefore finding $V(s)$.

Let's say, that agents only information is a sequence of rewards he receives from executing different actions in different states. In this case, agent can only attempt to learn numerical evaluation of reward function in different states. This corresponds to choosing a set of heuristic rules, that from agent's experience maximize his reward, but he does not have complete knowledge of why those rules work for him. A more rigorous definition of such learning process follows.

It is possible to re-define $V(s)$ in more general form. Let's define a function $Q(s, a)$ as expected reward from executing action a in a state s and following optimal policy afterward.

$$Q(s, a) = E[R_a(s, s') + \beta V(s')]$$

This definition is referred to as *Q-function*. This function is useful because if agent learns values of *Q-function* for different actions in different states, he will be able to find optimal policy without explicit knowledge of underlying rewards $R_a(s, s')$ and probabilities $P_a(s, s')$. This follows from the fact, that $Q - function$ is exactly the quantity that is maximized in the definition of $V(s)$. Therefore, for the optimal action

a^* in current state following holds:

$$a^* = \arg\max_a Q(s,a)$$

So, by learning *Q-function* can can learn $V(s)$ and therefore optimal policy without any knowledge of $R_a(s,s\prime)$ and $P_a(s,s\prime)$. Notice also this relation between the Q-function and the value function:

$$V(s) = \max_a Q(s,a)$$

Therefore, we can introduce a recursive definition of *Q-function*, not unsimilar to *Bellman's equation*. This definition is at the core of algorithm, that will enable the agent to learn *optimal policy*.

$$Q(s,a) = E[R_a(s,s\prime) + \beta \sum_{s\prime \in \mathcal{S}} P_a(s,s\prime) Q(s\prime,a')]$$

4.2 Q-learning algorithm

As it is so with $V(s)$ function, agent does not know it's values for particular actions in particular states. Her/his only option is to make some estimate through iterative approximation. From now on, agent's estimate about the true value of $Q(s,a)$ will be denoted $\hat{Q}(s,a)$.

Iterative approximations of $Q(s,a)$ and therefore *learning* the optimal policy is described by the following algorithm, which is in the literature referred to as *Q-learning algorithm*.

For each s, a initialize the table entry $\hat{Q}(s,a)$ to zero.
Observe current state s
Do forever:

- Let n be the number of current iteration
- Select an action a and execute it
- Receive immediate reward r
- Observer the new state $s\prime$
- Update the table entry for $\hat{Q}(s,a)$ as follows:

$$\hat{Q}_n(s,a) \leftarrow (1-\alpha_n)\hat{Q}_{n-1}(s,a) + \alpha_n[r + \beta \max_{a'} \hat{Q}_{n-1}(s\prime,a')]$$

where

$$\alpha_n = \frac{1}{1 + visits_n(s, a)}$$

and *visits* stands for total number of times that the $state - action$ pair has been visited up to the current iteration.

- $s \leftarrow s'$

Following (Mitchell 1997), we can introduce a theorem, that guarantees the model agents to eventually learn *optimal policy* and therefore act as agents in RBC model presented in Galí 2008 would.

Theorem (Dayan 1992)

Consider a Q-learning agent in a stochastic Markov Decision process with bounded rewards $(\forall s, a)|r(s, a)| \leq c$. *The Q learning agent uses the training rule of Equation (eq2), initializes its table* $\hat{Q}(s, a)$ *to arbitrary finite values, and uses a discount factor* β *such that* $0 \leq \beta < 1$. *Let* $n(i, s, a)$ *be the iteration corresponding to ith time that the action* a *is applied to state* s. *If each state-action pair is visited infinitely often,* $0 \leq \alpha_n < 1$ *and*

$$\sum_{i=1}^{\infty} \alpha_n(i, s, a) = \infty, \quad \sum_{i=1}^{\infty} [\alpha_n(i, s, a)]^2 < \infty$$

then for all s *and* a, $\hat{Q}(s, a) \rightarrow Q(s, a)$ *as* $n \rightarrow \infty$.

For this theorem to hold, standard dynamic macroeconomic assumption are sufficient, but strong-Form Rational expectations are not required. Notice however, that the Q-learning algorithm as presented above does not specify, how action a is chosen in each round and that for the *theorem 1* to hold, each state-action pair needs to be visited infinitely often. Therefore, simply choosing the most "natural" procedure of action selection - choosing the $action$ that is associated with highest value of $\hat{Q}$ won't be enough. This problem can be solved by using so-called "ϵ - greedy Q-learning" (as presented in Sinitskaya 2015). This variant of Q-learning introduces some $\epsilon \in (0, 1)$. Decision maker with probability ϵ chooses random action $a \in A_s$ (where $A_s \subset A$ is a subset of possible actions in current state s) and with probability $1 - \epsilon$ chooses action $a \in A_s$ which is associated with maximum $\hat{Q}$. This ensures, that as $n \rightarrow \infty$, each $state - action$ pair is visited infinitely often.
This theorem therefore requires weak-Form rational expectations extended for people's willingness to experiment every now and then. Of course, in this particular model setting, this assumption may be unrealistic. Decision, that is picked with probability ϵ is completely random. This is not very common behavior of real human decision makers, and certainly will cause some level of noise in model results. This level of noise can be adjusted by setting reasonable values of ϵ. Also, people decision making may sometimes indeed seem random - especially when they have very limited amount of information about their environment. (Kahneman 1974)

4.3 Adaptive Q-learning (with learning rate)

Previously introduced *Q-learning* algorithm guarantees a convergence of Q value estimates in Markov decision process. However, in some cases it has significant drawback. Agents using this algorithm are not able to react to changes in the distribution of stochastic variables in the process. Or to put it another words, once they learn something, they cannot start to learn something new. If for example at time $t = 5000$ agent has some reasonable estimate $\hat{Q}(s,a)$ of the true $Q(s,a)$ and exogenous change to the environment causes this estimate to be no longer valid, agent can not react. The *visits* counter is likely to be too large to allow for any significant change in $\hat{Q}$.
In theory, such changes should not happen. The state space $\mathcal{S}$ should contain every single one possible state of the world imaginable and no changes should be allowed and in theoretical work, this is very reasonable. But in real application of *Q-learning* such exogenous changes might happen. One of the variants of *Q-learning* solves this problem by not using *visits* counter and using some constant $l \in (0,1)$ called *learning rate* instead. So the complete *ϵ-greedy Q-learning* algorithm with *learning rate* is described as follows:

For each s, a initialize the table entry $\hat{Q}(s,a)$ to zero.
Observe current state s
Do forever:

- Let n be the number of current iteration
- Select an action a and execute it
- Receive immediate reward r
- Observer the new state $s\prime$
- Update the table entry for $\hat{Q}(s,a)$ as follows:

  $$\hat{Q}_n(s,a) \leftarrow (1-l)\hat{Q}_{n-1}(s,a) + l[r + \beta \max_{a'} \hat{Q}_{n-1}(s\prime, a')]$$

  where

  $$l \in (0,1)$$

  l is a parameter, often referred to as *learning rate*, which enables the agent to adapt in changes in state transition probabilities and reward functions.

- $s \leftarrow s\prime$

Disadvantage of this approach is that convergence theorem presented above no longer holds. Advantageous is however the ability of agents to adapt to some technological growth process when using this algorithm. This reason led to selecting this variant of the algorithm for the final macroeconomic model.

5 Games of multiple learning agents

In previous sections, concept of reinforcement learning was introduced, with the focus on particular reinforcement learning algorithm. Agents in the model will use this algorithm for their decision making. Such model, with multiple interacting entities that react to each other, should be put into some context of previous research. First, we will consider concept of learning automata, original developed by M. L. Tsetlin in USSR during 1960's. In the introduction of learning automata I will follow Narendra and Thathachar 1989. Then, a general concept of games of multiple agents will be introduced. Section *N-player stochastic games* will define theoretical concepts needed not only to define the model this thesis develops, but also to examine its behavior. Key game theory terms, such as *Nash equilibria*, need to have a rigorous definition in the context of previous section on *Q-learning* and *Dynamic programming.*

5.1 Learning automata

Consider an *adaptive decision-making entity*, whose key characteristics is, that at any given time t, the entity is in some well-defined state. The entity is in some environment, from which it can take inputs. Also, the entity can influence its environment back, through some outputs.
In this sub-section, and in this sub-section only, following notation applies:

- Φ is a state of the entity
- β is an input from the environment to entity
- α is output from the entity to the environment

Set of states in which the entity can be is finite and can change over time. Importantly the state does not evolve just by itself but every change of the state of the entity is a result of some input from its environment.
To make the concept of learning automata more understandable for the reader, let's look at the three words *adaptive decision-making entity* more closely.
Adaptive means that there is some function $\mathcal{F}$ that maps current state Φ and input into the next state of the automaton.

$$\mathcal{F} : \Phi \times \beta \rightarrow \Phi$$

Decision-making means, that there is some function $\mathcal{H}$ that maps the current state and input onto the output:

$$\mathcal{H} : \Phi \times \beta \rightarrow \alpha$$

By the word *entity* we mean, that there exists some record of a current state $\Phi(n)$.

To summarize, learning automaton is a quintuple:

$$< \Phi, \alpha, \beta, \mathcal{F}, \mathcal{H} >$$

Automata can be both stochastic or deterministic, depending on the properties of funtions $\mathcal{H}$ and $\mathcal{F}$. Basic automaton, or so called finite state-action automaton is defined as follows. (Narendra and Thathachar 1989)

1. The state of the automaton at any instant n, denoted by $\phi(n)$, is an element of the finite set
$$\Phi = \{\phi_1, \phi_2, \ldots, \phi_s\}$$

2. The output or action of an automaton at the instant n, denoted by $\alpha(n)$, is an element of the finite set
$$\alpha = \{\alpha_1, \alpha_2, \ldots, \alpha_r\}$$

3. The input of an automaton at the instant n, denoted by $\beta(n)$, is an element of a set β. This set could be either a finite set or an infinite set, such as an interval on the real line. Thus
$$\beta = \{\beta_1, \beta_2, \ldots, \beta_m\} \; or \; \beta = \{(a, b)\}$$
where a and b are real numbers.

4. The transition function $\mathcal{F}$ determines the state at the instant $(n + 1)$ in terms of the state and input at the instant n
$$\phi(n + 1) = \mathcal{F}[\phi(n), \beta(n)]$$
or $\mathcal{F}$ is a mapping from $\Phi \times \beta \rightarrow \Phi$ and could be either deterministic or stochastic

5. The output function $\mathcal{G}$ determines the output of the automaton at any instant n in terms of the state at that instant
$$\alpha(n) = \mathcal{G}[\phi(n)]$$
or $\mathcal{G}$ is a mapping $\Phi \rightarrow \alpha$ and is again either deterministic or stochastic.

Definitions above imply, that automaton is able to generate s sequence of outputs. In case of stochastic automaton, recall that this means that $\mathcal{G}$ is stochastic, each output $\alpha_i, i = 1, 2, \ldots, r$ is executed at any n with certain probability p_i. So action α_i is executed at time n with probability $p_i(n)$. Note not only the indexing for elements of α but also the period indexing (n). This is because those probabilities can change over time, depending on current state $\Phi(n)$. There is a type of automaton for which the state Φ is defined as a set of probabilities $p_i, i = 1, 2, \ldots, r$ for which following holds:

$$\sum_{i=1}^{r} p_i = 1$$

This is very useful concept, because automaton can adapt to its environment by changing the values of particular probabilities.
Before we define more on adaptive automata, we will need some comparison, or a benchmark automaton. A pure chance automaton is automaton that has the same probability for executing all outputs:

$$p_i(n) = \frac{1}{r}$$

This automaton does not adapt to it's environment and its action are driven by pure chance. So any automaton that could be considered adaptive, or learning, needs to perform better.
Meaning of *perform better* also needs definition. Let's say, that we would want the automaton to solve some problem. To produce a series of output that would have some desired outcome. If for example our automaton would be a household in economic model, we might want it to maximize its utility gains. Or if the automaton would be a car driver traveling to a destination, we might want it to use the quickest path.

5.1.1 Penalty probabilities

In order to be as general as possible, the meaning of *perform better* is put in a following way. Consider a stationary random environment in which automaton operates. Automaton can take some action out of α. Taking such action can result in penalty. Penalty is something to be avoided, whatever it actually might be. Penalty is form of response of the environment and therefore $\beta = \{0, 1\}$. This means that penalty either happened or did not happen. With each member of α is associated a probability $c_i, i = 1, 2, \ldots, r$ of penalty, or in another word of $\beta(n) = 1$. Quantity, representing average penalty for state $\Phi(n)$, or set of probabilities $p_i(n)$ if you will is denoted

$$M(n) = E[\beta(n)|p(n)] = Pr[\beta(n) = 1|p(n)] = = \sum_{i=1}^{r} c_i p_i(n)$$

For a pure chance automaton

$$M(n) = \frac{1}{r} \sum_{i=1}^{r} c_i$$

Automaton a that is said to learn, must be at least better than pure chance automaton.

A learning automaton is said to be *optimal* if

$$\lim_{n \to \infty} E[M(n)] = c_l$$

where

$$c_l = \min c_i$$

5.1.2 Variable structure automata

Previously, we have defined a *Q-learning* algorithm. Entity using this algorithm can be viewed as something in literature called variable structure automaton. Theory of automata precedes $Q-learning$ but are the same in many ways. For a *ϵ-greedy Q-learning* there is a set of actions like the set α. Also, given some input, there is a vector of probabilities that the entity will take some action. With *ϵ--greedy Q-learning* action with the highest associated $\hat{Q}$ is taken with probability $1-\epsilon$. All other actions are taken with probability $\frac{\epsilon}{r}$.
The $\hat{Q}(s, \alpha)$ updating rule defined previously is a form of $\mathcal{F}$.
Generally, there are multiple types of $\mathcal{F}$ and in the literature on learning automata $\mathcal{F}$ is much more simpler that *Q-learning*, which goes beyond some constraints in the definition of automata.

5.1.3 Reinforcement schemes

In automata theory, the term *Reinforcement schemes* is used for methods of updating probabilities p_i. Several such methods have been developed over the years. Key properties of such schemes are convergence and optimality.
Optimal action in period (n) will be the one with the lowest associated probability of penalty. A schema is said to be optimal is if it assigns lowest values of p_i to such actions. In order to do that, computing probabilities c_i is necessary, no matter whether implicitly or explicitly. in order for the scheme to find optimal set of values

of p_i, it is necessary to execute each action infinite number of times because only then the estimates of c_i will converge to their true values.
Similar thing can be seen in *Q-learning* convergence theorem.

5.1.4 Games of interconnected automata

Imagine following situation. Each day, multiple car drivers need to travel from one place to another. There are always multiple routes they can take. More drivers can take same route which will result in denser traffic in particular spaces which slows them down. Driver's - Automaton's target is to find a route, that will take least amount of time.
If there would be only one traveling automaton, using some appropriate learning method and multiple trials would eventualy lead to a traveling scheme that would require the least time.
The situation would be very different in case of multiple automata game. During first few trials, some automata might discover the best route and use it to their benefit. But as more and more automta discover this route to be the quickest, dense traffic might actually make this route much slower. Thus, some other route might be quickest now. And automata might switch from one route to another. But on that new route same thing might happen as well.

Systems like this have interesting properties. For example, traffic might after many trials spread out in such way, that all possible routes might take the same time and therefore all automata might be indifferent between switching routes. In this particular case, system would reach a *Nash equilibrium*.

Behavior of a collective of automata can be more complex than simple sum of of behavior of its members. Since such properties emerge in economic model this thesis develops, following section takes a more closer look on games of multiple learning entities, including a definition of *Nash equilibrium* for such systems.

5.1.5 Agents in economic model as learning automata

In this section, a theoretical concept of learning automata was introduced. Main, and for this thesis crucial, element of learning automata theory is that there is some current *state* of the automaton, some record of its current properties at any given time, that can be used for distinguishing automata. If our automata would be for example firms and households in economic model, we could distinguish them based on their state.
Let me finish this section with important statement that is the main reason why I bothered the reader with section on learning automata in the first place.

It should be reasonable to assume by now, that in a game of automata, that are the same in terms of state spaces Φ, β, α and function $\mathcal{F}$ and $\mathcal{H}$, the current states of all automta $\Phi(n)$ form *Markov chain*. Therefore, if their environment is a *Markov chain*, then the overall state of the game is a *Markov chain* as well. More specifically, a *Continuous state Markov chain*, since p_i is not countable, but from interval $(0, 1)$. This has great importance when considering possible convergence properties of whole game.

5.2 N-player stochastic games

In this section, a general concept of *n-player stochastic games* is introduced. Definitions that will follow are taken from Schwartz 2014. Notation was altered whenever it was necessary to match concepts and definitions from previous section on *Q-learning*.
Theory of learning automta does not provide sufficient framework to study games of interacting agents, or economic models. More generality is necessary and some key *game theory* definitions. These are regularly utilized in economics as well and are needed for the model this thesis develops to be well.
The agent based macroeconomic model in this thesis could be defined as a *n-player stochastic game*

Definition 1. *An n-player stochastic game Γ is a tuple $<\mathcal{S}, \mathcal{A}_1, \dots, \mathcal{A}_n, R_1, \dots., R_n, P, \beta>$, where $\mathcal{S}$ is the state space, $\mathcal{A}_i$ is the action space for agent $i (i = 1, \dots, n)$, $R_i : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N \rightarrow R$ is the reward function for agent i, $P : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N \rightarrow \delta(\mathcal{S})$ is the transition probability map, where $\delta(\mathcal{S})$ is the set of probability distributions over state space $\mathcal{S}$ and $\beta \in (0, 1)$ is a discount factor.*

The state space in case of macroeconomic *game* could be compromised of possible values of goods and labor prices. Or in a simpler game it could be a finite and enumerable set.

Definition 2. *Joint action set is defined as*

$$\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N$$

Joint action set is a set of all possible action that all players (agents) can take. In a macroeconomic model, such set is a combination of all possible values of labor, consumption level and number of bonds of each agent. Or again, it could be finite and enumerable set in case of a simpler game.
Sets of action can form joint policies. Recall that policy is an ordered sequence of actions given some sequence of states. Joint set of policies can form a *Nash equlibrium*.

Since *Nash equlibrium* is a state in which no agent can gain anything from changing it's policy, prior to definition of *Nash equlibrium* it self, we need to establish what it means to *gain*.

Definition 3. *Discounted sum of rewards for agent $i(i = 1, \dots,)$ in game Γ*

$$R_i(s, \pi_1, \dots, \pi_n) = \sum_{t=t_0}^{\infty} \sum_{s' \in \mathcal{S}} \beta^{t-t_0} R_{a_t}(s, s'|\pi_1, \dots, \pi_n) * P_{a_t}(s, s'|\pi_1, \dots, \pi_n)$$

So no agent can in *Nash equlibrium* increase its discounted sum of rewards by altering its policy:

Definition 4. *In stochastic game Γ, a Nash equilibrium point is a tuple of n policies $(\pi_1^*, \dots, \pi_n^*)$ such sthat for all $s \in \mathcal{S}$ and $i = 1, \dots, n$,*

$$R_i(s, \pi_1^*, \dots, \pi_n^*) \geq R_i(s, \pi_1^*, \dots, \pi_{i-1}^*, \pi_i, \pi_{i+1}^* \dots, \pi_n^*)$$

Or in another words, this definition can be summarized a follows: Nash equilibrium is a state of the system, in which set of all policies, that all agents use for their decision making, are optimal for them, given that all other agents use their corresponding optimal policies.

5.2.1 Player dependence on other players actions

In and stochastic game, in opposite to simple Markov decision process, players action, together with the environment does not completely determine the state transition or reward. Look at the probability transition matrix and reward functions for n-player stochastic games:

$$P : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N \rightarrow \delta(\mathcal{S})$$

$$R_i : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N \rightarrow R$$

The game may be *played* by players whose decisions are determined by a adaptive decision making mechanism. If the adaptive decision making mechanism observes only the current state of the game and rewards, the dependance among players will bring make a interesting phenomenon arise.
At least one player not using *Nash equlibrium* policy will cause other player's *Nash equlibrium* policies not to yield the highest payoff. Therefore more player may then diverge from *Nash equlibrium*. In this

The dependence of agents on one another actions may prevent the system in such situation from ever reaching *Nash equlibria* or it may slow down possible convergence to *Nash equlibria* as shown later in this work as a phenomenon called *Long preserving* *shock*. This however does not apply to a decision making mechanism that take into account not only the state and reward, but the actions taken by other player as well. These mechanism try to predict the behavior of other players and then modify the policy accordingly. One such adaptive mechanism is introduced in the next section.

5.2.2 Nash Q-learning

There is a variant of *Q-learning* algorithm, called *Nash Q-learning*, that is worth looking at. Although this particular algorithm is not utilized in the models this thesis presents, the reader of this thesis should be introduced into this method anyway. This sections loosely follows Hu and Wellman 2003.

In order to use *Nash Q-learning* in a model, additional assumption has to be made: Each player in game Γ has information about actions taken by other players, as well as the rewards they have received. On top of that, simulation using this algorithm requires more computational power. Those were the reasons why this algorithm was not selected. It has however more advantageous convergence properties (Convergence to Nash equilibrium).(Schwartz 2014)
Basic idea behind this algorithm is, that each player keeps track of actions take by all other players and rewards they have received. Using this knowledge, each player tries to estimate so called *NashQ* function. *NashQ* is defined as a discounted sum of rewards for a player when all players follow their *Nash equilibrium* strategies from next round onward.
Nash Q-learning algorithm is summarized as follows: Initialize table of $\hat{Q}^j(s, a_1, \dots, a_n)$ where $j = 1, \dots, n$ and n is a number of agents in the game.
Important note: This means, that each agent keeps tract of all estimates $\hat{Q}(s, a_1, \dots, a_n)$ not only for itself but for every other agent as well. Also, the estimates are not just for particular player actions, but for set of actions selected by all player at particular rounds.
Do forever:

- Let t be the number of current iteration
- Select an action a and execute it
- Receive immediate reward R
- Observer the new state $s\prime$ as well as $R^j; j = 1, \dots, n$ and $a^j; j = 1, \dots, n$ where n is number of agents in game, R^i is reward *j-th* agent and a^i is action executed by *j-th* agent

- Update the table entry for all $j = 1, \dots, n$ $\hat{Q}^j(s, a)$ as follows:

$$\hat{Q}_t^j(s, a_1, \dots, a_n) \leftarrow (1 - l)\hat{Q}_{t-1}^j(s, a_1, \dots, a_n) + l[r + \beta NashQ_t^j(s\prime)]$$

  where

$$l \in (0, 1)$$

  l is a *learning rate*

- $s \leftarrow s\prime$

Precise calculation of $NashQ_t^j(s\prime)$ is done by each player based on their estimates $\hat{Q}^j(s, a_1, \dots, a_n)$ using quadratic programing. (Schwartz 2014)

6 Reinforcement learning in stochastic games

Before a economic model based on concepts presented so far was developed, a simpler simulation was used to test the properties of system of two iterating *Q-learning* agents. On a smaller scale, property that can be seen in economic model as well emerged already and rigorous definition of such behavior is therefore provided later in this section.

6.1 An example of multi-agent game: Football

Imagine a football team that plays infinite number of matches against its opponents. This team has two strikes, called "Player 1" and "Player 2". Both player play not only against opposing teams, but since they are footballers, they compete against one another for fame as well.
Each game ends by the same situation. This is not very realistic, but just imagine that it is indeed the like so. The score is even and the game heads to a draw. In the last minute, the two strikers take possession of the ball and haste towards the opposing team's net. If they manage to score, the game is won, because there is not enough time left for the other team to score as well.
Initially, Player 1 has the ball and Player 2 runs slightly ahead of him. On the diagram below, This means that Player 1 is in position 1 and Player 2 is in position 2.

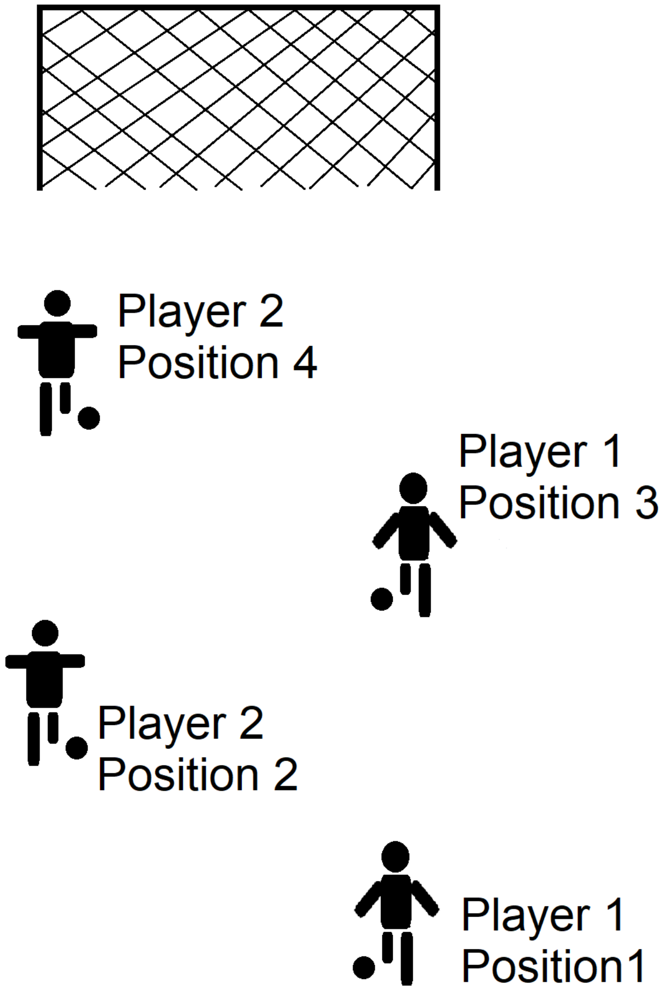

Figure 2: Football game

Player 1 can not run any further with the ball. Opposing team defense men would take the ball away from him. At this stage he has only two options. He can either shoot at the net, or pass the ball to Player 2 and then run further to position 3. If he shoots and scores a goal, not only his team wins, but he will also become fans hero, since he is the one who decided the match. Unfortunately, the net is still quite far away and shooting in position 1 would give him only a little chance of scoring. Perhaps, passing the ball to Player 2, who is closer to the net is a better option.
Let's say that he indeed did pass the ball to Player 2. Player 2 now has the ball and again has only two options what to do. Shooting at net or passing the ball back to Player 1, who in the meantime ran forward and is now in position 3. His chances

of scoring are now better than what Player 1's in position 1, but still quite not high enough. Logical choice would be to pass the ball back.
But this is a football and for a particular player, not only the outcome of the match matters. Personal gain, in form of the fame for deciding the match is worth a lot as well. If he passes the ball, his team's chance to win is going to improve greatly. But his chances of deciding the match drop. Player 1 might not be willing to pass the ball back again, once Player 2 runs into Position 4. Position 4 is indeed directly in front of the net, and scoring a goal from position 4 is certain. But Position 3 is good as well. Player 1 is highly likely to try his chances from this position and become the hero of the match himself. So when Player 2 is in possession of the ball in position 2, he has to weigh shooting and passing against one another carefully.
Now, both player use *ϵ-greedy Q-learning* and the game is played infinite number of times. Players have no explicit knowledge of scoring probabilities from different positions. They have to implicitly calculate them. Only thing they can observe is the outcome of the game. Whether they have received a payoff for wining the game and whether they have received the playoff for deciding the match which is significantly larger, than just for winning.

Following table presents the chances of scoring from different positions.

Table 1: Chance of scoring

Position 1	1 %
Position 2	20 %
Position 3	50 %
Position 4	100 %

Simulation was coded in Matlab R2018a and ran. Clear strategy (policy) for both players emerged quickly and formed a Nash equilibrium. Player 1 in position 1, having almost no chance of scoring always passes the ball to player 2. But player 2 in position 2 always shoots. So on average, the team wins only 1 out of 5 matches. Player 2 lowered the chances of the team of wining because the fame for deciding the match was high enough. And player 1 still passes even though he never gets the ball back. Position 1 is simply too bad to score. This is a *Nash equilibrium*, but it is nowhere near pareto-optimality.
Now lets imagine that there is sudden exogenous change of the players environment. Let's say that the team coach have had enough of wasted games and introduces a fine for each player that shoots but misses in position in which passing was better option for the team. Also, he does not tell the players anything in advance.
In the matlab simulation, there is a trigger that changes the environment in certain period. to reflect the angry coach. Also, the state action pair *visits* counter in *Q-*

learning algorithm gets zeroed. Figure of Q estimates development below, shows what happened.

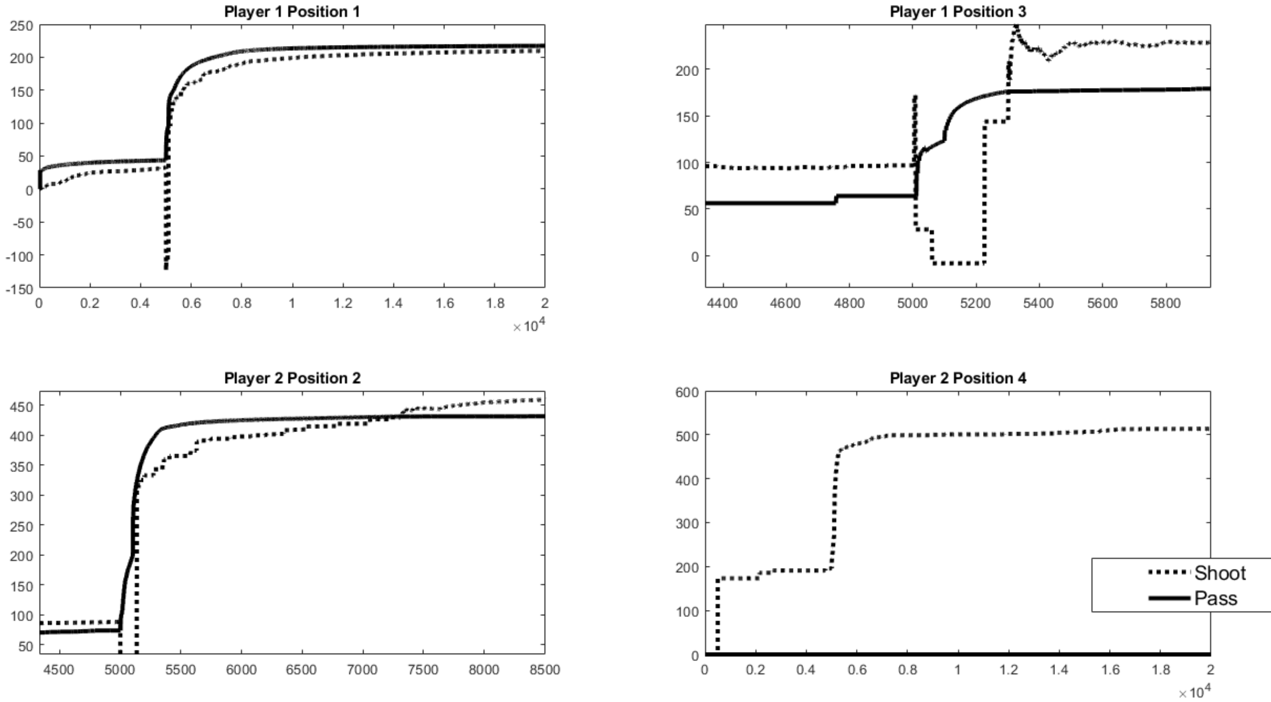

Figure 3: Q-values after shock

At first, the estimate of payoff for shooting for Player 2 in position 2 drops sharply. Now he always passes the ball, 20% is simply too low chance of scoring in presence of fines for missing the net. But look at player 1 in position 3. Previously he would shoot if he got the chance of having the ball in position 3, but the fine is still high enough for him to play it safe and pass the ball back. Now the team wins every game because position 4 guarantees certainty of scoring. This is also *Nash equilibrium*, but this time also *pareto-optimal*.
But now the environment changes again. The coach is happy with the wins and decides to cancel the fines. Again, without telling the players a word. So the fines were in effect only for a short time, from $t = 5000$ to $t = 5100$
As we examine what happens after the fines are canceled and the game is in it's initial setting, interesting phenomena emerges. Again remember, that the *Nash equlibrium* is now player 1 passing the ball in position 1 and Player 1 shooting tha ball in position 2. But even after period $t = 5100$, the Player 1 still passes the ball back to Player

2. That is because *ϵ-greedy Q-learning* needs some time to adopt, and the smaller ϵ the longer it will on average take for the player 1 to try to shoot and discover that it actually brings larger payoff on average. Let's call this state of the game a shock a provide it with the following definition:

Definition 5. *Let π_i^t be a policy that is used by agent i in a period t of stochastic game Γ. A Shock is a period of length $k \in \mathcal{N}$ during which policies $(\pi_1^t, \ldots, \pi_n^t)$ for all $t = t_0, \ldots, t_0 + k$ do not form a Nash equilibrium if $(\pi_1^t, \ldots, \pi_n^t)$ form Nash equilibrium for $t = t_0 - 1$ and $t = t_0 + k + 1$*

But as long as player 1 will bass the ball back from position 3, player 1 has no reason *not* to pass the ball in position 2. That would actually lower his chances of scoring. So even if he would by chance try the his *Nash equilibrium* policy, he would not stay with it for long. His optimal policy is to pass in position 2 for as long as he keeps getting the ball back.

This means, that for the whole game it takes longer on average to reach *Nash equilibrium* if the players interact. And by interact I mean that action of one influences the other as well. Only when player 1 stops passing the ball back, which happened around $t = 5250$, the player 2 has the chance to actually start learning his *Nash equilibrium* policy as well. Therefore, the shocks in games of interacting agents, that influence one another take longer on average compared to games of agents that can act independently. In this example game of football, the length of chain of dependency is equal to the number of agents in the game. In other games, chain of dependency can actually by shorter that the number of agents in the game. But as longs as there are agents whose decisions depend from one another, shocks will be longer. Let's define this phenomena and call it *Long preserving shock*.

Definition 6. *In stochastic game Γ, in which N interconnected agents interact and use use ϵ-greedy Q-learning algorithm for their decision making, a long preserving shock is a shock for which on average $k \geq \frac{1}{\epsilon^N}$.*

This definition is important in economics. When learning entities interact, such as it happens in economy, exogenous changes of environment cause a period of instability. This period gets longer the more entities depend on one another decisions sequentially.

Note that in real economy, there is significant amount of information sharing. Information sharing would shorten such shock or maybe prevent *Long preserving shocks* altogether.

Also, in case the word *interconnected* in this definition would be ambiguous: Interconnected in this instance means, that players actions directly influence one another's rewards.

7 Simple Economy without money market

To examine the behavior of the economy compromised of Q-learning households and firms, a simplified variant of Classical monetary model is considered. These simplifications are quite major and are described in following paragraphs. Q-learning algorithm works better with fewer possible states and actions. Since money market, in the form of specified in budget constraint in Galí 2008 model adds complexity. Price of bonds Q, which is a state variable, must have at least some values. Model without money market will be simpler and households and firms will learn policies more quickly – it will take less time for them to utilize all possible actions in all possible states. If state space is constructed only by W x P, it has significantly fewer elements as state space W x P x Q. Another reason for omitting money market in the simplest model is, that every time household wants to sell bonds, there must be someone who would buy them. Otherwise Q could not be determined. So, if we omit bonds, only remaining state variables from classical monetary model are price of goods P and wage W. Therefore defining the state space $\mathcal{S}$ is done by defining sets W and P. State space is common for both households and firms.
Control variable for households are labor N and consumption C.

7.1 Omitting expectations causes Uroboros problem

As previously stated, P, W and Q form a state space for households and firms in the classical monetary model. When one wants to develop an agent based model utilizing reinforcement learning algorithms and base the model on the classical monetary model (or any other DSGE model) following issue emerges.
At time t, the exact value of P_t, W_t and Q_t needs to be determined. This process seems straight forward. Let's look at the determination of P_t. It is assumed, that the economy is in equilibrium, i. e. supply of goods is equal to the demand for goods. This makes determination of price dynamics and therefore P_t possible. In order to find the equilibrium value of P_t, aggregate supply and aggregate demand in the economy need to be known. Both aggregates are determined by the individual demand and supply. And the individual demand and supply is given by particular consumers and firms action, represented by the value of control variables C and N. These values (or actions) are chosen as follows:

1. Take subset of all possible $Q(s,a)$ given particular s.
2. $a = \arg\max(Q(s,a), a \in A)$

The whole process is depicted in picture. Equilibrium price forms a state. State, serving as an input for Q-learning algorithm, determines each individual firms and households action. Combined actions form aggregate supply and demand. Aggregate supply and demand form equilibrium price – or in another words, state. This forms an infinite loop, much like Uroboros.

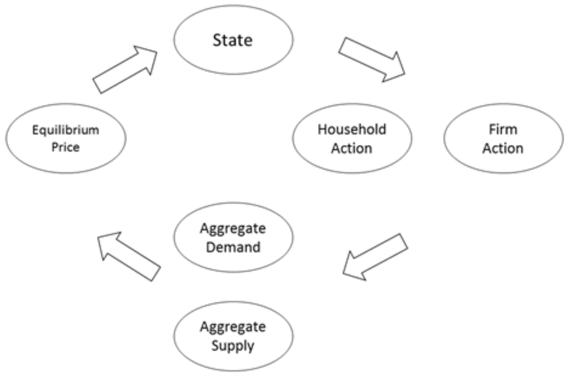

Figure 4: Loop

There are multiple ways how to approach this. Apart from those commonly used in economics, this thesis provides another two options that are more suitable for multi-agent model. But first, let's look at this from the real world perspective. Level of consumption of a real-world household is given by the relation between its subjective marginal utility and real price level, meaning the ration of wage and price of goods. In larger contracts, like the purchase of a house or a car, household can negotiate the price. In retail, like buying a bar of chocolate, they take the price as given. It is set by the seller. In many cultures, even the retail prices are subject to negotiations. Take a Middle East market, the buyer and the seller negotiate the price. In western world, the price is also a subject to negotiations. But these negations take place between retailer and its suppliers and can be readjusted only occasionally because readjusting prices is costly itself. The negotiating positions are defined by complex analyses of their costs and markets that they operate on. One household does not have enough power to even enter into negotiations and therefore takes the price as given. Simplest answer to the question how does the price emerge would be, that it develops over time. It does not emerge at tome time t and at time t+1 would be readjusted.

7.2 Divine "mindreading" auctioneer

The method for determining the equilibrium P, W and possibly Q at each t used in this thesis is a specific form of Walrasian auction. In general, by Walrasian auction is considered following process introduced by Léon Walras and could be considered a simplest solution to above stated problem. In Walrasian auction, each agent in the economy has a knowledge of his demand for good at each price or can calculate his demand for each price. Note that this applies to the Q-learning agents in this model as well. This will be addressed in more detail later. Above that, there is a divine auctioneer in the economy, with the power to set the price. He does this in the following manner. Each agent in the economy (both firms and households) submit their supply and demand bids for each price. The auctioneer then compares the total amount supplied and demanded for each P and selects the price for which supply is equal to the demand. This way, Walras's law is satisfied, as excess supply and demand sum to zero. Walras proposed a so called tâtonnement process through which equilibrium price is to be achieved. This is a process of trial end error. At the beginning, the auctioneer sets a random price. Then agents submit their demand and supply bids for that price. If supply is lower then dement, the auctioneer increases the price and vice versa. In case of discrete time in which price could be readjusted only once every turn in would take some time for the price to converge to equilibrium. Thus, Walrasian auction is usually assumed to take place simultaneously and auctioneer is assumed to be "divine" in the sense, that he possesses perfect information. Equilibrium can be therefore achieved immediately. In the multi-agent Q-learning based model it is possible to introduce such auctioneer. With every agent there is a set of estimates of Q function values for each state and action. So, it is possible to obtain the information about individual supply and demand for each price level. The auctioneer can simply have all the estimates of Q of all agents in the economy. Therefore, it has the information about individual supply and demand levels. It can search through all values of P and set the price level in a such way, that the excess supply and demand are equal to zero, or at least as close to zero as possible. It is reasonable to note at this point, that equilibrium with zero excess supply and demand may not be always achievable. The model economy is discrete not only in time, but also in quantities and prices and this property may cause, that supply simply cannot equal the demand in some cases.

There are other methods how to describe a process of reaching equilibrium.

7.3 A complete description of model economy

A model economy is based on classical monetary model but has significant differences. Time t is a discrete variable. Economy consists of finite number of agents. There are two types of agents: households and firms. Each time t firms hire labor and use their production function to produce goods. The production function will be specified later. Firms seek the optimal amount of labor to hire each t to maximize discounted sum of profits for from each period. At the end of each period, each firm's profits are distributed to households equally. Households sell their labor to firms each t in exchange for wage. They use the wage to buy goods produced by firms. For the sake of simplicity, it is assumed that there is only one type of good. The model economy has the property, that all contract are settle at the end of period. Both households and firms use epsilon-greedy Q-learning algorithm for their decision making. Besides households and firms, there is one more entity. A divine walrasian auctioneer whose purpose is to designate the state of the world. Auctioneers objective is (in this order of priority) to set the price of goods so that the goods market clears or that the difference between supply and demand is as small as possible, set the wage that the labor market clears or that the difference between supply and demand is as small as possible, and to minimize the inter-temporal difference in goods price and wage. State of the economy is defined by the price of goods P and wage W. Price can take following values:

$$P = \{1, 2, 3, 4, 5\}$$

And wage can take following values:

$$W = \{1, 2, 3, 4, 5\}$$

Therefore, there are 25 possible states of the model economy. Each of them will be examined by the auctioneer to determine equilibrium state for each period. Households offer their labor to the firms in exchange for wage and buy goods for a unit price. Households are wage and price takers. They cannot influence the price set by the auctioneer directly. Each period they choose an action, which is an amount of labor they offer to firms at given wage and amount of goods they consume at given price. Amount of labor can take following values:

$$N = \{0, 1\}$$

Simply put, they can either participate in the labor market or not. Level of consumption can take following values:

$$C = \{0.5, 1\}$$

Level of consumption is connected to their current money balance. They can have spent half of their budget, retaining the rest of budget to the next period, or entire budget on purchasing consumption goods. This ensures the budget condition from original monetary model is retained and limits the number of possible action so reinforcement algorithm can work more efficiently. At each period, firms hire labor first, and used the labor to produce goods they sell to customers. They are as well as households price and wage takers. Their objective is to maximize discounted sum of profits Level of labor hired by firm can take following values:

$$N = \{2, 3, 4, 5\}$$

With the restriction, that they must be able to cover the wages by selling the goods: P*N*A-N*W>=0. So, the values of N that would violate this condition are automatically removed from the list for the period. To sum up, each period following events take place in this order:

1. Divine auctioneer calculates excess goods demand and supply and excess labor demand and supply. Calculation is done by comparing Q function estimates of each agent for each possible state (combination of values of P and W), selecting the highest Q estimate for each agent and obtaining his amount demanded. This can be understood as if auctioneer would be asking each agent about his bids for goods and labor given the price and wage levels. Auctioneer then aggregates goods demand and supply as well as labor demand and supply for each P and W combination and selects the combination of values that will minimize the difference between goods demand and supply. At the end of this process, the state of the model economy is determined.

2. Households select their action – Values of C and N based on epsilon-greedy Q-learning algorithm. Firms select the amount of labor demanded using their epsilon-greedy Q-learning algorithm.

3. Labor is allocated. Since the amount of labor supplied and demanded are natural numbers, a situation in which supply is not equal to demand even though auctioneer tried to match those can arise. In that case, a random firms or households, depending on whether the excess is on the demand or supply side are selected. The selected agents will not be able to participate in the labor market in this period and therefore obtain wage or sell goods.

4. Goods are exchanged for money. Again, since the amount of goods supplied and demanded are natural numbers, a situation in which supply is not equal to demand even though auctioneer tried to match those can arise. In that case, a random firms or households, depending on whether the excess is on the demand or supply side are selected. The selected agents will not be able to participate in the goods market.

5. Firms update their Q-function value estimates. Then their profits are discarded. No distribution of profits to households was implemented.

6. Households consume goods, gain utility and update their Q-function values estimates. There is one more thing to note: Epsilon-greedy variant of Q-learning algorithm may cause the agent to select random action (amount of labor and goods) instead of the one with the highest associated Q-function value estimate. This would result in excess in demand or supply because the auctioneer

would have wrong information about the agent's bid. So, to avoid this, establishment of random action selection for each agent is done prior to event 1. In another words, for each agent the random action selection rule is evaluated prior to all other events. Even before the state of the world is determined. Then, a random action is assigned to each possible state of the world and agent must take this action and provide the information about all randomly selected actions to the auctioneer as if they would be the ones with highest Q-function values estimates.

7.3.1 Note on *Long preserving shocks*

Wide spread of information that is accurate and that agents believe to be truth and update their policy accordingly leads to shortening of shocks.

7.4 Equilibrium determination

Previously selected mechanism of determining the state s is based on review of different equilibrium determining processes. In short, aggregate demand and supply is computed for each $s \in \int$ based on values of $\hat{Q}(s, a)$. The state for which the deference between supply and demand is the lowest is selected as the nest state. Economic theory provides several mechanisms, or processes, by which equilibrium can be determined. This section provides review of three processes that were considered during the model development along with explanation on why the simplest *Tâtonnement process* was ultimately selected for usage.

7.4.1 Engeworth process

Egenworth process, or Engeworth's theorem, describes how equilibrium price can emerge in competitive market. Suppose there are two agents, A and B, and two goods. Each agent is endowed with either good 1 or good 2. Both agents have concave indifference curves. Agents can exchange the goods at no charge to increase their overall utility. In the two-agent case, it can be shown that there exists so called exchange path, which is a line that connects two points on both agents indifference curves and defines a set of points on which both agents would be better of if they would trade. Reader can image this as two charts two-dimensional charts (one dimensions for each good) with two sets of indifference curves facing each other. The equilibrium path connects the two indifference curves that represent the level of goods in the economy. Line connecting the two curves is called contract path. Equilibrium lies on the contract path, but it is not possible to determine where exactly. Such graph is called Engeworth box. Let's say, that two new agents emerge in the

economy. Each of them is identical to one of the agents that are already present. This will cause the exchange path to shorten. If trade would to occur between two agents at either starting point or end of the exchange path, one of the agents would receive all the benefits from the trade. The agent that received less from the trade (the minimum that would make him better of than not trading) now can trade with his identical counterpart. Since they are identical, they will split the goods. The third agent, who has not traded yet now must offer those two a better deal, than his identical counterpart did in the first trade, because both are now on higher indifference curve. This effectively shortened the exchange path. Adding more agents into the economy shortens the exchange path further. In the case of perfect competition, this would result in one equilibrium point being achieved. Such games, or processes were later thoroughly investigated by Shapley and Shubik 1971. In their model of two-sided market concerned indivisible units of good, such as houses, each participant either supplied or demanded exactly one unit. They utilized linear programing to solve the model. Linear programing comes as a natural, when analyzing optimization problems defined by multiple inequalities. This is a clever approach but is not very suitable to be utilized in the solving of the problem proposed above and therefore for this thesis.
As presented above, the state $\int$ could not be in a agent-based model selected at once. It would have to be sequentially determined as more and more agent wold be added. This would create large complexity in the model coding, which would go beyond scope of this thesis. Also, *Q-learning* cannot handle the negotiations between two agents and therefore could not determine where on the trade curve would for example first two agents trade.

7.4.2 Hahn process

Another example of process that would lead to some equilibrium price is so called "Hahn process" after Frank H. Hahn, developed by Hahn and Negishi 1962. As summarized by Neill and Boland 1986, Hahn process is characterized as follows. Let's suppose that there is a market for some good i at some price p, that might be out of equilibrium. Trading at that price will always lead to one of two following results:

- If good I was in excess demand before trading, then after trading there will be done, no market participant will hold more of good i than he desires to hold.

- If good I was in excess supply before trading, then after trading there will be done, no market participant will hold less of good I than he desires.

After trading is done, the price of good I will rise if the good i was in excess demand, or fall, if the good was in excess supply.

Fisher 1983 later adds to this another assumption: Market is well organized and buyers and sellers can trade relatively quickly to the rate at which the price is adjusted. Which shall make Hahn process somewhat less restrictive with weaker assumption compared to Endgeworth process.
Endgeworth process could be implementer relatively easily into the model, but the speed of change of prices relative to the speed at which agents would be able to trade wold have to determined somehow. This would require additional assumption in caparison to ultimately chosen solution. The rate of change would have great effect on the overall model results since agents are reactionary as well due to the nature of *Q-learning*.

7.4.3 Tâtonnement process

Let's suppose that there is a market for good i. On this market, some number of agents or traders operate. Some supply the good, some demand the good. Also, there is an artificial entity, often referred to as "divine auctioneer" or "walrasian auctioneer". The auctioneer quotes some price for good I and all agents submit their supply and demand bits. But no trade takes place currently. Auctioneer simply aggregates supply and demand and calculates the difference. If there is excess supply, auctioneer calls out a price that is lower. If there is excess demand, auctioneer calls out a price that is higher. This process is repeated until supply of good i equals the demand for that good. (Takayama 1985)
This market clearing mechanism can be implementer into agent-based model with *Q-learning* with relative ease and requires no addition assumptions. Only assumption being that markets indeed do clear. It was selected as a state determination mechanism for the final model.

8 Model results

Model presented in previous sections was coded in Matlab, specifically Matlab R2018a was used. Initial goal of using C++ was omitted due to the number of modification and testing that needed to be done. Matlab made changing the model and testing in much easier. Also, even in Matlab the simulation was reasonably fast. Usually, 5000 rounds with 1200 agents in the simulation could be finished in about an hour running Matlab on Dell Latitude E6410 laptop with Intel Core i7 processor and 8 GB of memory.

8.1 Production function

In classical monetary model, firms use a production function that assigns a level of output to a level of labor and technology. technology process is usually assumed to be a random variable with some trend in it. Randomness was for a sake of simplicity omitted and following production function was used:

$$Y_t = round[(e^{N_t/1.2})^{0.8}(1+0.01t)^{1-0.8}]$$

Therefore, the exogenous technology process is

$$A_t = 1 + 0.01t$$

This is not a stochastic process, but it easily could be. *Q-learning* should handle it with no problem, as long as the distribution remains constant in parameters.

8.2 Random queue clearing mechanism

As shown previously, a general equilibrium theory was used for determination of market clearing wages and goods prices. Namely, divine Walrasian auctioneer looks into the $\hat{Q}$ tables for each agent and set the price and wage so that the difference in supply and demand in the goods market is as small as possible.
But there would always be some difference and also we cannot forget the labor market, on which the difference between supply and demand might be even more significant. This is in the model handled as follows:
Each period, auctioneer sets the price and wage. Then, using matlab's pseudo-random numbers generator households and firms are ordered into randomly ordered queues. First, labor is considered. First firm demanding labor is assigned a first household willing to supply labor, if firm's labor supply is not satisfied, it is assigned next household until firms demanding labor, or households supplying labor run out. This results in randomly selected firms or households to be left out of the labor market.
Then, total goods production is calculated. Households demanding goods are again randomly ordered in a queue as well as firms that produced goods. Households goods demand are satisfied in this random order until either goods, or goods demanding households run out. Again, this lefts randomly selected firms with unsold goods or households with unsatisfied demand out of the market.

8.3 Initial model run

Model was ran for 5000 rounds, therefore $t = 1, \dots, 5000$, with quite substantial amount of firms and households. Exact model settings are presented in the table:

Table 2: Model Parameters

Variable	Value
firms	200
households	1000
P	$\{1, 2, 3, 4, 5\}$
W	$\{2.2, 4.4, 6.6, 8.8, 11\}$
C	$\{0.5, 1\}$
N-households	$\{0, 1\}$
N-firms	$\{0, 2, 3, 4, 5\}$
β	0.95
ϵ	0.1
l	1/3

Each period, data on goods produced by firms and unemployment rate was collected. Note that unemployment rate is a percentage of households willing to work and not being able to. It is NOT a percentage of all not working households. Households that removed them self from the labor market in particular period by not supplying labor are not included in unemployment rate calculation.
Collected data are presented in a figure:

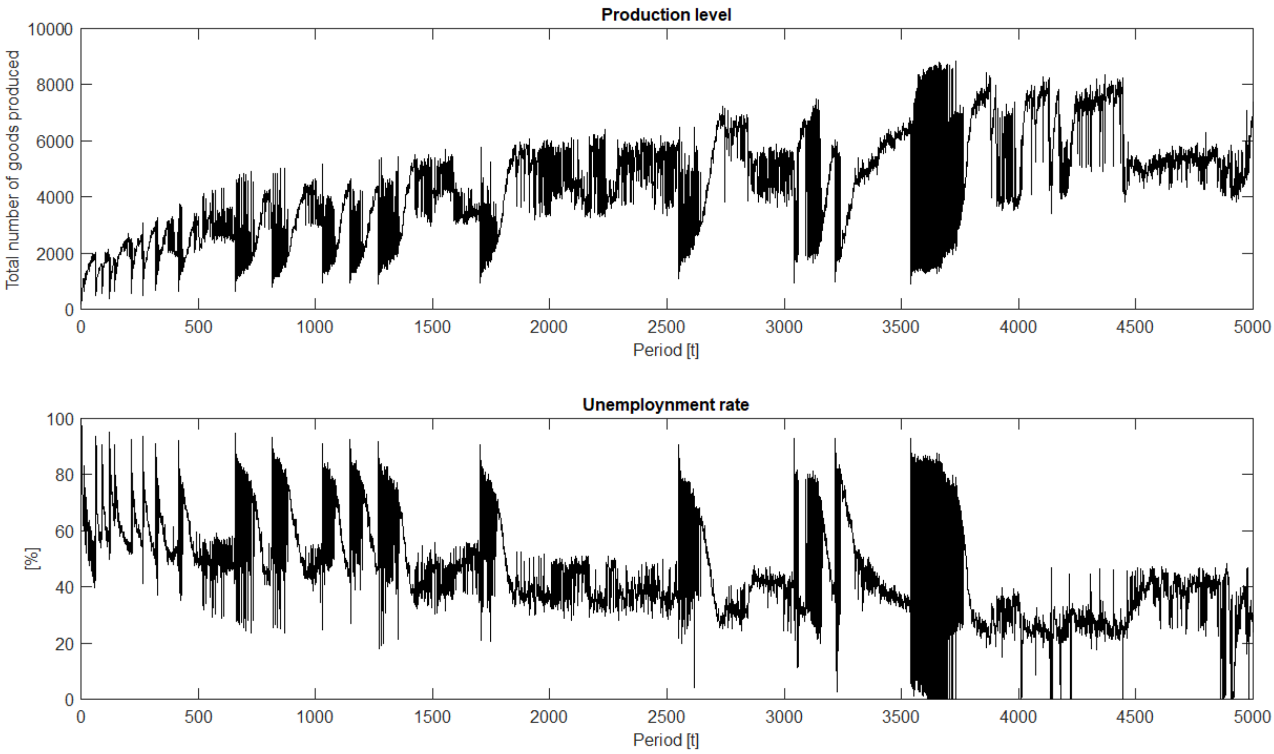

Figure 5: Initial model run

First thing we can notice is, that production level is cyclical. Apart from smaller volatility, caused by Random queue clearing mechanism, there are periods when consumption rises and then suddenly drops. Unemployment rate works the same but in the opposite direction. Each cycle is characterized by unemployment slowly decreasing only to experience sudden increase at the beginning of the next cycle.
Cycle are caused by changes in price/wage levels. Every time new combination of price and wage gets set by the auctioneer, firms and households need to learn their optimal policies for this particular combination. As they learn, the production arises and unemployment falls. Not only the production is rising, but production as well. They might, and in this particular model setting they are, rising at a different pace. So the goods supply/demand difference is slowly increasing as the agents in learn their optimal policies.
Once the supply/demand difference reaches certain level, auctioneer select a different price/wage combination and whole learning process starts over.
Overall growing trend in consumption is caused by the increasing A_t and simulates

real economy growth. This trend is purely exogenous.
Learning process, and therefore consumption climb and unemployment fall is quite slow. Slower than ϵ setting might suggest. This has the same cause as *Long preserving shock* has. Take for example some price and wage level set by the auctioneer. There is some amount of labor that each firm should hire in order to maximize its profit and some *Nash equilibrium* strategy for all firms. At the same time it might be optimal for all households to offer their labor at the labor market. Again a *Nash equilibrium* strategy for households.
Firm will have a probability ϵ that is selects a random action each round. If the firm by chance select the optimal amount of labor to hire, one would expect the firm to obtain high profit and update it's estimate $\hat{Q}$ of corresponding Q and thus hire same profit maximizing amount of labor in next periods as well. Once it executes the *Nash equilibrium* action it should follow the *Nash equilibrium* policy from now on. But what might happen in this model and what actually contributes to the slower pace of production growth after price/wage change is similar phenomenon that we could observer in the game of football example. The firm might select the *Nash equilibrium* action but it might produce very low profit because it for example won be able to hire the amount of labor it demands. Simply because many households did not learn their *Nash equilibrium* strategy yet. And until they do, the firm might not be able to learn it either.
After the simulation was finished, $\hat{Q}$ for all agents was used to construct goods supply and demand curves for all real price levels.

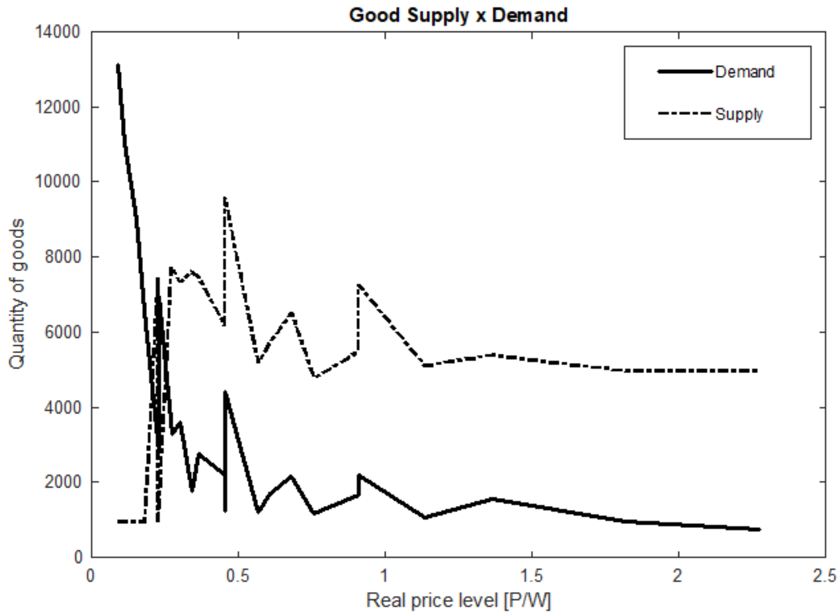

Figure 6: Goods market

This diagram makes one thing very clear. Values of $\mathcal{S}$ do not enable markets to clear given the number of firms, households and properties of the production function. Therefore, based on this observation, $\mathcal{S}$ was adjusted in the following way:

Table 3: Model Parameters

Variable	Value
firms	200
households	1000
P	$\{1, 2, 3, 4, 5\}$
W	$\{4.5, 9, 13.5, 18, 22.5\}$
C	$\{0.5, 1\}$
N-households	$\{0, 1\}$
N-firms	$\{0, 2, 3, 4, 5\}$
β	0.95
ϵ	0.1
l	1/3

Possible wage levels were increased, giving the households larger purchasing power and making hiring labor for firms relatively more expensive. All other parameters were kept the same. Again, same data was collected and is presented in a figure:

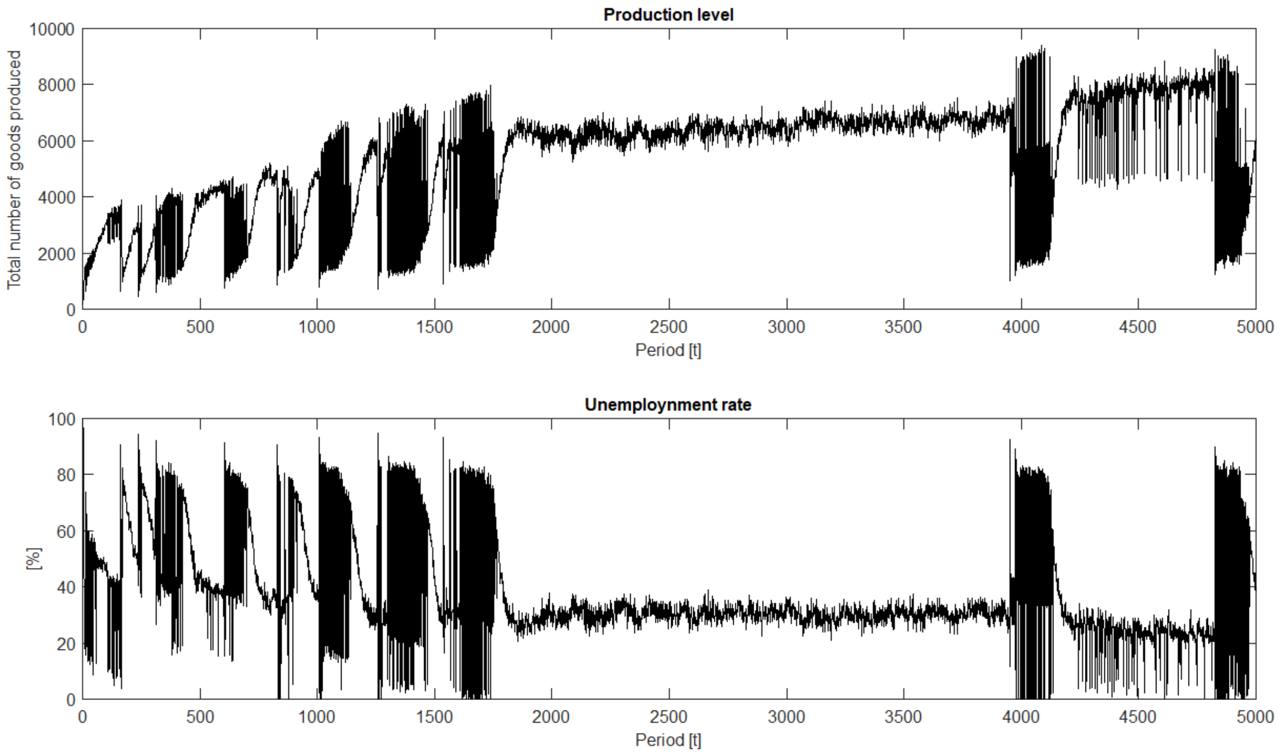

Figure 7: Adjusted model run

It can be seen, that this time model reach some stable state after period t = 2000. Price and wage levels remained constant. But then again cyclical fluctuation returned. Supply/demand curves after period $t = 5000$ were again constructed:

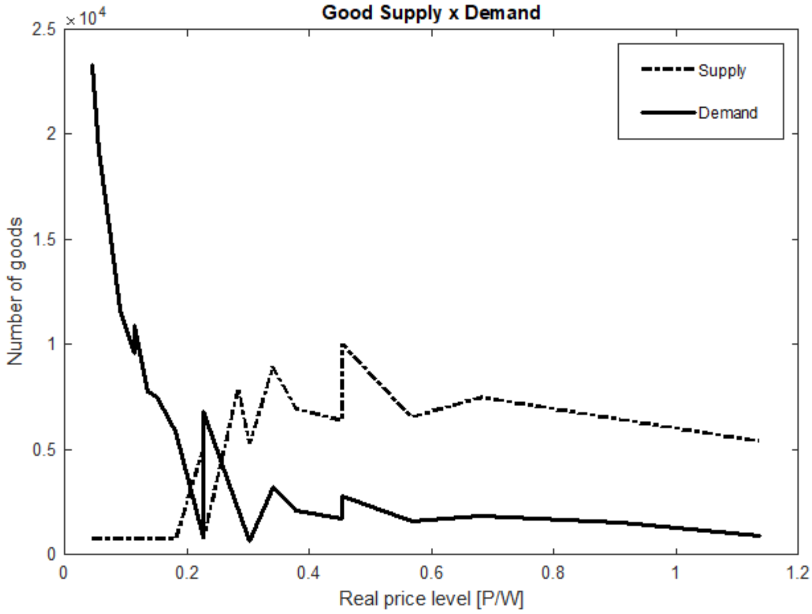

Figure 8: Goods market

And again, after examination, more adjustments were made to possible price and wage level values to make more room for reaching equilibrium at the end of the simulation.

Table 4: Model Parameters

Variable	Value
firms	200
households	1000
P	$\{1, 1.1059, 1.2117, 1.3176, 1.4235\}$
W	$\{5.0839, 5.2018, 5.3197, 5.4377, 5.5556\}$
C	$\{0.5, 1\}$
N-households	$\{0, 1\}$
N-firms	$\{0, 2, 3, 4, 5\}$
β	0.95
ϵ	0.1
l	1/3

In this model run, one more interesting thing can be seen. Since price and wage levels should be more "on spot" given the production function and number of firms and households.

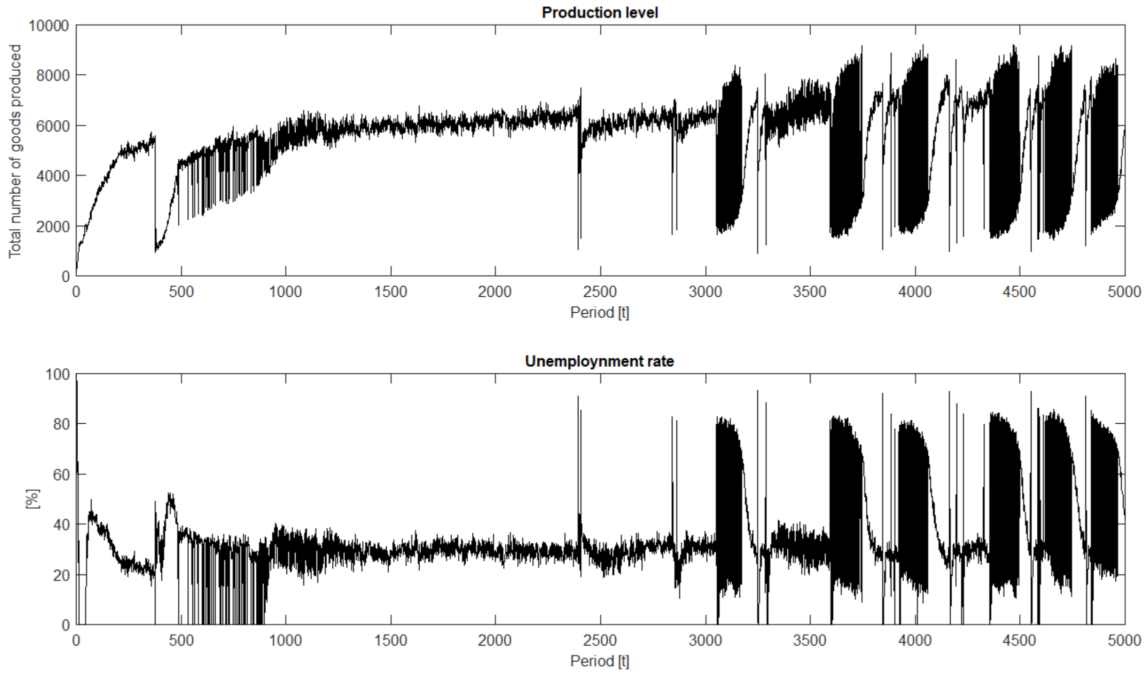

Figure 9: Model with close to equilibrium price and wage levels

After initial period of searching for market clearing wages and prices, economy stayed stable, because this price/wage combination resulted in small enough difference in supply and demand. But after some time, namely about $t = 3200$, "crises" again emerged. At this point, all agents likely learn their optimal policies given the state, but auctioneer still had to readjust price/wage level. This may be because of the growth in technology which was approaching the technology levels for which supply demand curves were constructed. Conditions simply changed and therefore re-adjustment was necessary. This hypothesis was examined more thoroughly in the next section, with different model setting. Namely by removal of technology growth for initial periods.

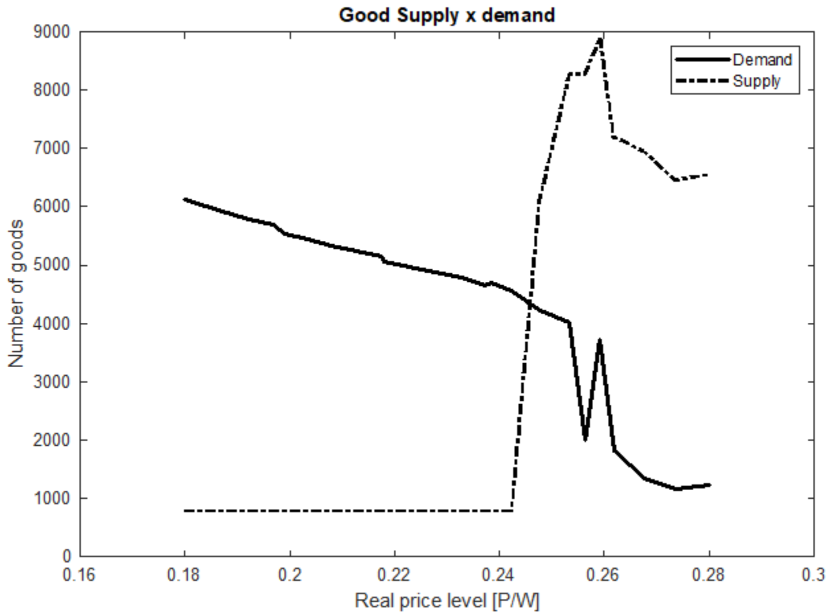

Figure 10: Goods market

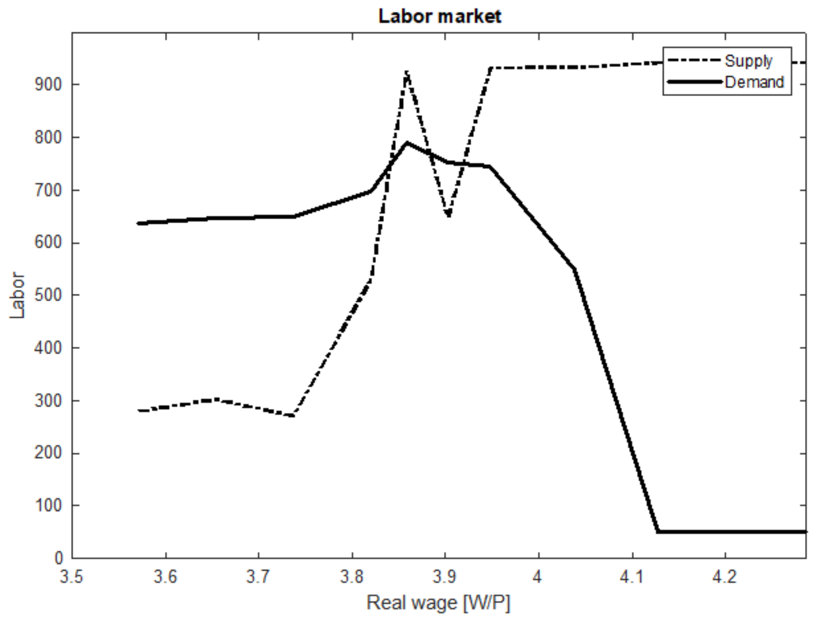

Figure 11: Labor market

For this third run not only goods demand/supply curve were constructed, but in

similar manner labor demand/supply was also examined.
Both goods and labor market curves look reasonable. Supply is more or less increasing with price rise and demand moves the opposite way around and there is some equilibrium level. The behavior of supply and demand can in this agent based model can be seen nicely on the labor market figure. First, the wage is discrete - perfect equilibrium can not be reached and this causes irregularities in the place where equilibrium would be. Second, even for very high wage to price ratio, the supply of labor was never full 1000 households. That is again because of the randomness in households decision making. From the figure we can see that maximum labor supply is approximately 950. Which is in line with 10% households selecting random action and therefore 5% households supplying no labor. Same applies to firms. Even at extremely high wage level, they will still demand some labor. But the supply and demand curves are flat for high wage levels.
Same thing applies to the goods market.

8.4 Growth as a source of crisis

In the final model setting I wanted to examine whether growth can indeed cause a crisis (fall in production, consumption and rise in unemployment rate). As I was assuming growth can a cause of some crises in the previous model runs. The simulation ran for 20000 rounds in total. For the first 10000 rounds, the production function remained constant. Therefore for each given amount of labor hired, the same amount of goods would be produces for all $t = 1, \ldots, 10000$.
Other parameters were set as follows:

Table 5: Model Parameters

Variable	Value
firms	200
households	1000
P	$\{1, 2, 3, 4, 5\}$
W	$\{2.2, 4.4, 6.6, 8.8, 11\}$
C	$\{0.5, 1\}$
N-households	$\{0, 1\}$
N-firms	$\{0, 2, 3, 4, 5\}$
β	0.95
ϵ	0.05
l	1/3

From the time series of consumption and unemployment rate, we can see, that the model reached some steady state:

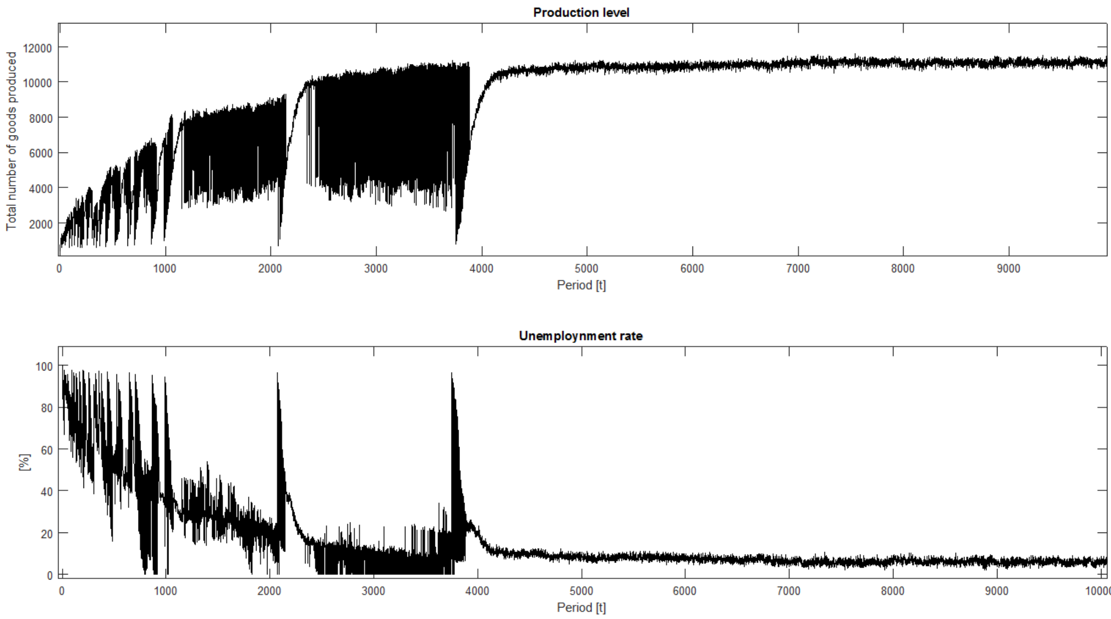

Figure 12: Consumption and unemployment level, run 4

After around $t = 5000$ no significant change in both consumption and unemployment rate can be seen. Steady unemployment rate was under 10% which can be explained quite simply by the effect of ϵ probability that at each round firms select the amount of labor hired by pure chance. This means that there will always by some level of unemployment in the model, depending on the value of ϵ. Let's call it a *natural level*.

Starting with the round $t = 10001$ a 4% exogenous technology growth was introduced into the model. Which had following results:

1. Per one unit of labor, more and more goods would be produced.

2. Households cannot increase their consumption. That is because no more labor can be hired by firms - unemployment is at natural level already. Therefore no more wages can be paid.

3. The difference between goods demand and supply starts to increase steadily.

4. At some point (to be specific at $t = 11799$) the difference is too large and price and wage level need to be adjusted, which causes a crisis.

Figure Below presents the values of aggregate production level and unemployment rate. A steady growth in production can be seen starting at the same time the 4% technology growth was introduced. The unemployment rate remains at the same constant level. (As well as consumption - not in figure) The crisis consists of sharp fall in production level and sharp increase in unemployment. As both firms and households adapt to a new state, production increases and returns to the original growth path. At the ent of the simulation, some volatility can be observed. This behavior becomes clear after we look at what happened during the firs crisis.

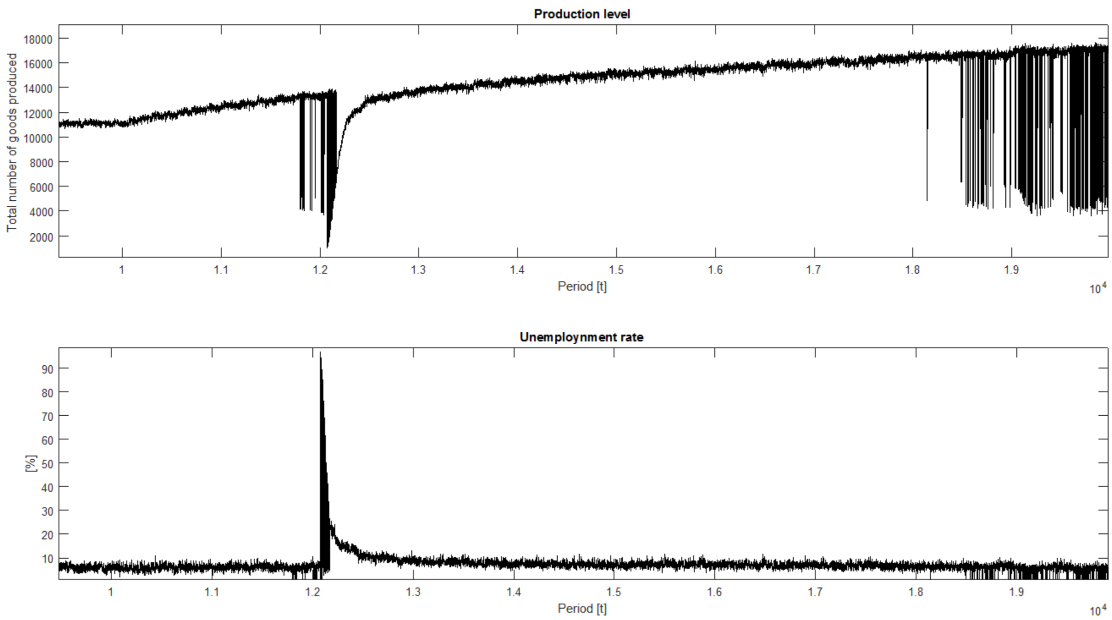

Figure 13: Consumption and unemployment level, run 4 - growth

Let's look at what happened after $t = 11799$ more closely. Following figure shows the price of good and wage as well. Several short, one period fluctuations in both can be observed. The level of consumption shows a constant value prior to the crisis, but production is in steady growth. Before the crisis, the equilibrium price and wage level was $P = 1$ and $W = 8.8$. So as the difference between consumption and production was increase it was inevitable, that at some point some other combination of price and wage would shirk this difference. And this is exactly what happened at $t = 11799$.

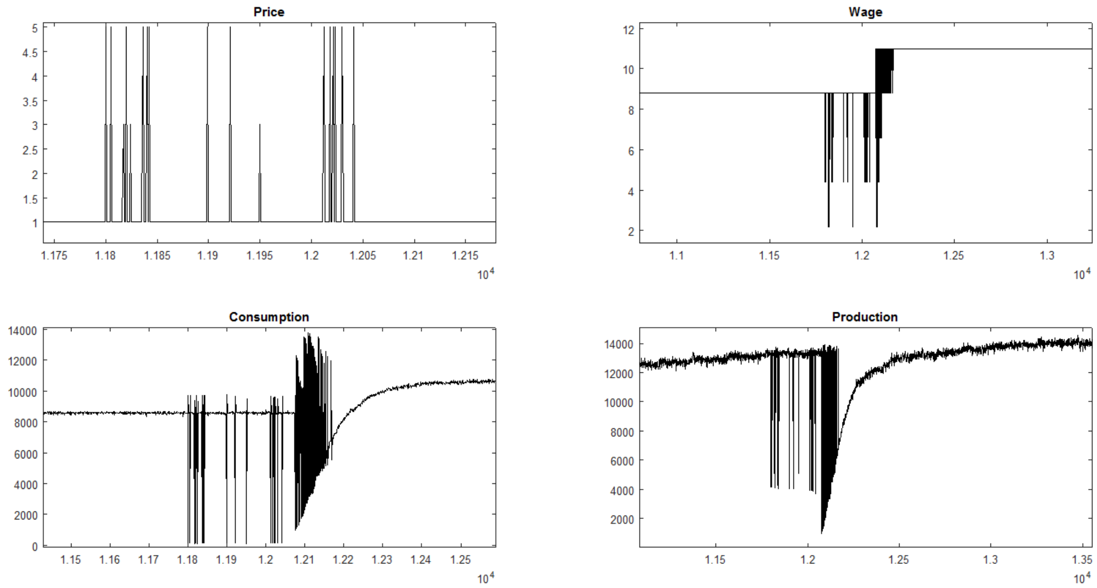

Figure 14: Crisis - detail

A different price of goods and wage was selected. Period of selecting different price and wage combination followed, which can be seen from the figure. Many different combinations of price and wage were selected in relatively short period of time, before the economy settled at some price and wage level. Each combination produced some supply/demand difference. Eventually, the the price wage combination for which the difference emerged the lowest was selected and economy returned to stable growth path again.
The reader may ask why the final combination of price and wage was not selected straight away, without the period of price and wage bouncing up and down as seen on the figure. This period of price/wage instability occurred because of the ϵ probability of agents selecting random actions. This property of *Q-learning* translates into each *state*, as represented by price and wage combination, produces some difference between supply and demand. And there is some randomness in this difference. The final combination of price and wage selected was $P = 1$ and $W = 11$. But the supply/demand difference for this combination might have been quite large before it was selected, because of this randomness, actually larger then for some other combinations. Also, if a combination of price and wage was visited by the economy in some previous period, agents might have already learned how to behave given such combination and therefor the difference might not have been so big. To sum up, given the nature of the firms and households decision making mechanism, a new

correct state needs to be visited several times. Before the economy can settle in it, a number of *wrong* states need to be visited and *eliminated*. Jumping in price and wage level causes also significant volatility in production and unemployment, which can be seen on the graph.
We can see, that after the economy settled, the price remained at $P = 1$, but the wage increased to $W = 11$. This is in line with what would one intuitively think should happen:**Technological growth should result in higher wage to goods price ratio.** And it indeed did. Once exogenous technology growth was introduced into the model, the ratio increased. Firms and households now have to adapt to a new state and learn the optimal amount of labor to supply and demand respectively. This period lasted for about 300 rounds during which consumption and production was increasing and eventually reaching stable level. The new stable consumption level is higher than the pre-crisis level. Therefore the goods supply/demand difference is lower and crisis ended. Production returned to the pre-crisis level and then continued on the technology growth path.

Now lets get back to examining the fluctuations in production and unemployment at the end of the simulation, starting around $t = 18145$. Starting from this point until the simulation ended, the production level as well as unemployment was jumping up and down. This is again because the technological growth caused the difference between goods supply and demand to be too large and the economy needed new equilibrium price and wage. At this point, based on the previous finding, new crisis should emerge. The crisis should start with period of extreme volatility(search for new equilibrium price/wage), followed by the period of rapid growth(agent adapting to the new state). But it did not. Instead the whole economy remained in the state of extreme volatility, with production jumping up and down. The cause of new crisis not emerging and economy ending just in this extremely volatile state is simple. Let's recall that in this particular model setting, possible values of price are $P = \{1, 2, 3, 4, 5\}$ and possible values of wage are $\{2.2, 4.4, 6.6, 8.8, 11\}$. And since after the last crisis economy settled in $P = 1$ and $W = 11$, the ratio of wage to price cannot be increased any further. Therefore it is not possible to reflect the growth in technology anymore. If it would be, a new crisis would emerge and it would eventually end just like the last one. But since it is not possible, the price and wage level will be jumping up and down, searching for a new equilibrium, possibly indefinitely.

9 Conclusion

This thesis developed macroeconomic Agent-based model, utilizing reinforcement learning algorithm. First, thesis shows that it is possible to define and code such

model and that a behavior of such model is reasonable and in line with intuition, when it comes to development of price, wage, production and unemployment. Several simplifications to the baseline textbook model had to be made. Whole money marked was omitted and all variables needed to be discretized. On the other hand, Perfect Foresight Strong-Form Rational Expectations assumption could be dropped. Also, firms and households decision making mechanism, while being very simple indeed, is not unnatural.
Simulation results show some interesting behavior as well. Model can not only reach some stable state, but when endogenous technology growth is introduced, crises can emerge. Thesis also explored the behavior of systems of interconnected agents, focusing on recovery from shocks.

9.1 Topics for further research

First thing one shall explore is adding money market to the economy. This would make experiments with monetary policy in the model possible. In could shows some nice results. During work on this thesis, simulation was ran dozens of times. Author was experimenting with price caps, minimal wage and so called *helicopter drop of money* which produced interesting results on the labor market.
More rigorous, or simply better definition of *Long preserving shock* is also something to be addressed.
Last but not least, author believes, that the discretization of all model variables is not necessary, maybe it could be addressed as well using fuzzy logic.

References

Bellman, R. E. (2010). *Dynamic Programig*. Princeton University Press.

Dayan, C. J. W. P. (1992). Technical note, q-learning. *Machine Leraning 8*, 279–292.

Fellous, R. D. S. M. J. F. J.-M. (2010). Computational models of reinforcement learning: the role of dopamine as a reward signal. *Cognitive Neurodynamics 4*, 91–105.

Fisher, F. M. (1974). The hahn process with firms but no production. *Econometrica 42*(3), 471–486.

Fisher, F. M. (1978). Quantity constraints, spillovers and the hahn process. *The Review of Economic Studies 45*(1), 19–31.

Fisher, F. M. (1983). *Disequilibrium Foundations of Equilibrium Economics*.

Galí, J. (2008). *Monetary Policy, Inflation, and the Business Cycle*. Princeton University Press.

Gilchrist, B. S. B. M. G. S. (1999). *Handbook of Macroeconomics*. Elsevier.

Hahn, F. H. and T. Negishi (1962). A theorem on non-tatonnement stability. *Econometrica 30*(3), 463.

Howitt, P. (2012). What have central bankers learned from modern macroeconomic theory? *Journal of Macroeconomics 34*, 11–22.

Hu, J. and M. P. Wellman (2003, 01). Nash q-learning for general-sum stochastic games. *Journal of Machine Learning Research 4*, 1039 – 1069.

Judd, L. T. K. (2006). *Handbook of Computational Economics, Volume 2*. North Holland.

Kahneman, A. T. D. (1974). Judgment under uncertainty: Heuristics and biases. *Science 185*, 1124–31.

Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.

Ljungberg, W. S. P. A. T. (1993). Responses of monkey dopamine neurons to reward and conditioned stimuli during successive steps of learning a delayed response task. *Journal of Neuroscience 13*, 900–913.

Marden, J., G. Arslan, and J. Shamma (2006, 01). Joint strategy fictitious play with inertia for potential games. *Automatic Control, IEEE Transactions on 54*, 6692 – 6697.

Marimon, R., E. McGrattan, and T. J. Sargent (1990). Money as a medium of exchange in an economy with artificially intelligent agents. *Journal of Economic Dynamics and Control 14*(2), 329 – 373. Special Issue on Computer Science and Economics.

Maršál, F. B. M. H. A. (2012). Survey of research on financial sector modeling within dsge models: What central bank learn from it. *Czech Journal of Economics and Finance 62*, 252–277.

Mitchell, T. (1997). *Machine Learning*. McGraw Hill.

Montague, W. S. P. D. P. R. (1997). A neural substrate of prediction and reward. *Science 275*, 1593 – 1599.

Moore, N. K. J. (1997). Credit cycles. *Journal of Political Economy 105*.

Narendra, K. and M. Thathachar (1989). *Learning automata: an introduction*. Prentice Hall.

Neill, R. and L. A. Boland (1986). *Methodology for a New Microeconomics*.

Niv, Y. (2009). Reinforcement learning in the brain. *Journal of Mathematical Psychology*.

Perkins, S. and D. Leslie (2014). Stochastic fictitious play with continuous action sets. *Journal of Economic Theory 152*, 179 – 213.

Riccetti, L., A. Russo, and M. Gallegati (2017). Financial regulation and endogenous macroeconomic crises. *Macroeconomic Dynamics*, 1–35.

Schwartz, H. M. (2014). *Multi-Agent Machine Learning: A Reinforcement Approach* (1st ed.). Wiley Publishing.

Shapley, L. S. and M. Shubik (1971). The assignment game i: The core. *International Journal of Game Theory 1*(1), 111–130.

Sinitskaya, L. T. E. (2015, December). Macroeconomies as contructively rational games. *Journal of Economic Dynamics and Control 61*, 152–182.

Takayama, A. (1985). *Mathematical Economics*. Cambridge University Press.

Tesfatsion, L. (2016). http://www2.econ.iastate.edu/tesfatsi/reintro.pdf.

Tversky, D. K. A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica 42*, 263–291.

Wang, Y. and L. Pavel (2014). A modified q-learning algorithm for potential games. *IFAC Proceedings Volumes 47*(3), 8710 – 8718. 19th IFAC World Congress.

Watkins, C. J. (1989). *Lerning from delayed rewards*. Ph. D. thesis, King's College.